

BFS Traversal

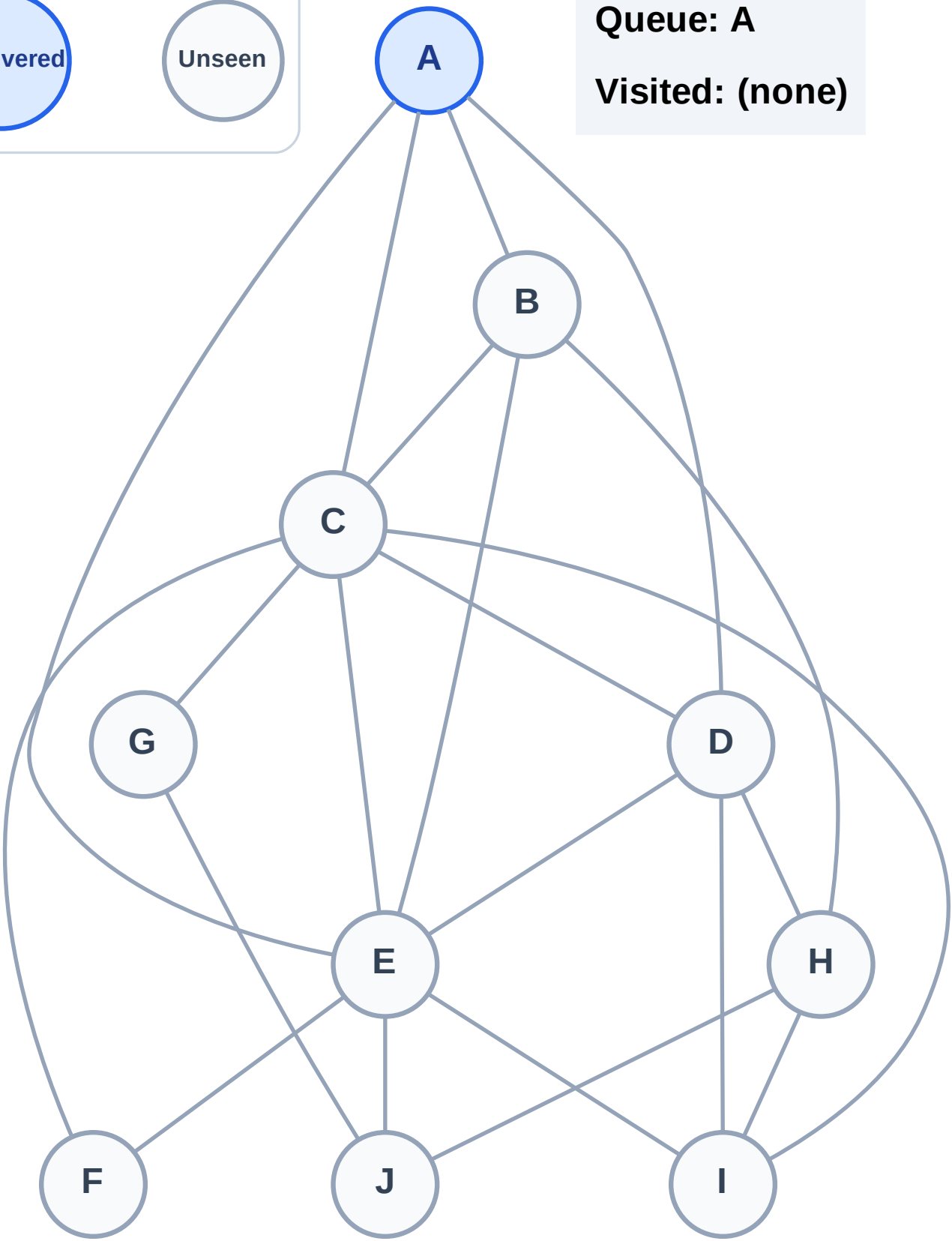
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

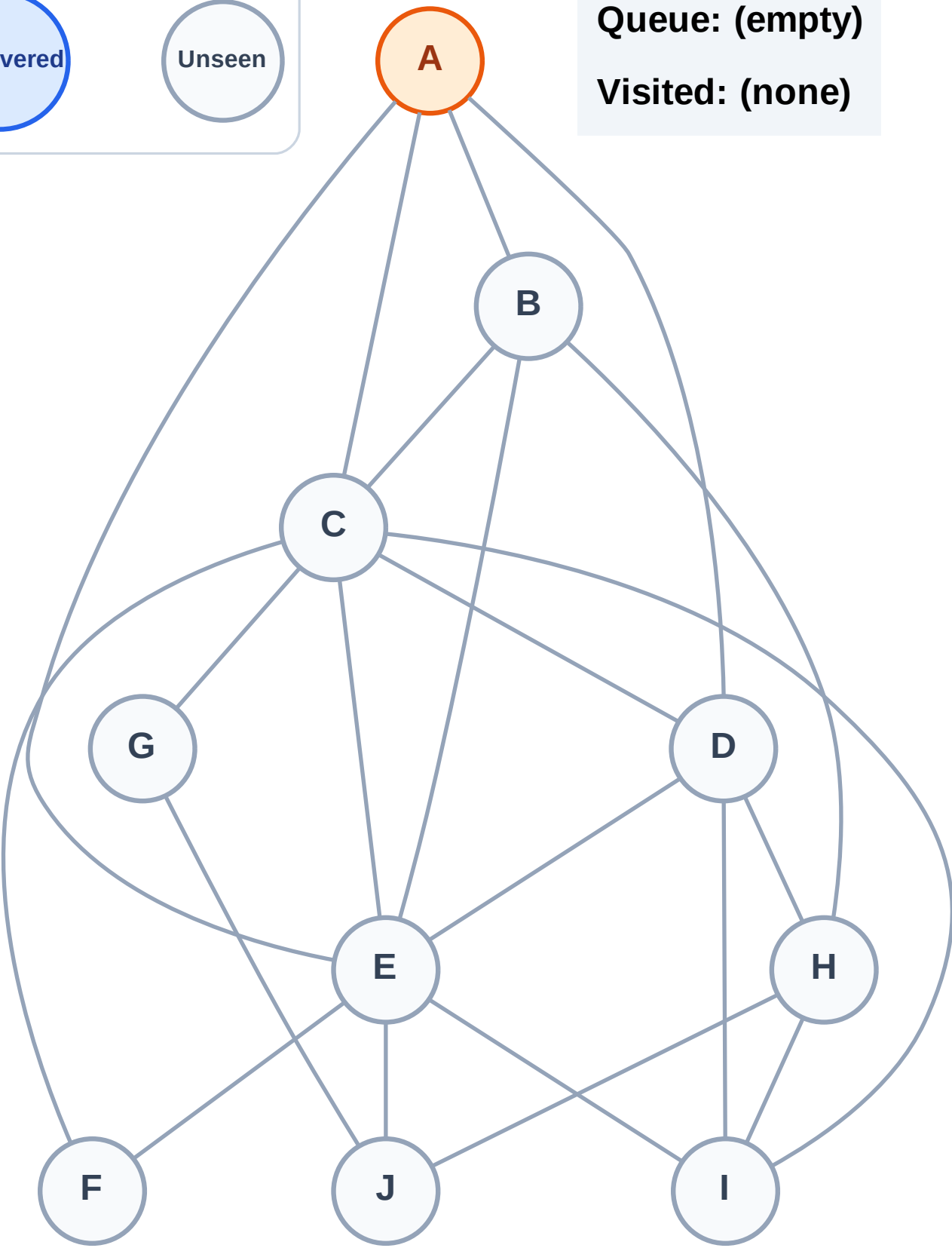
Complete

Processing

Discovered

Unseen

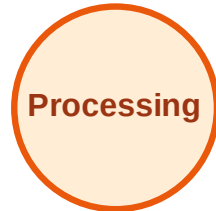
Queue: (empty)
Visited: (none)



BFS Traversal

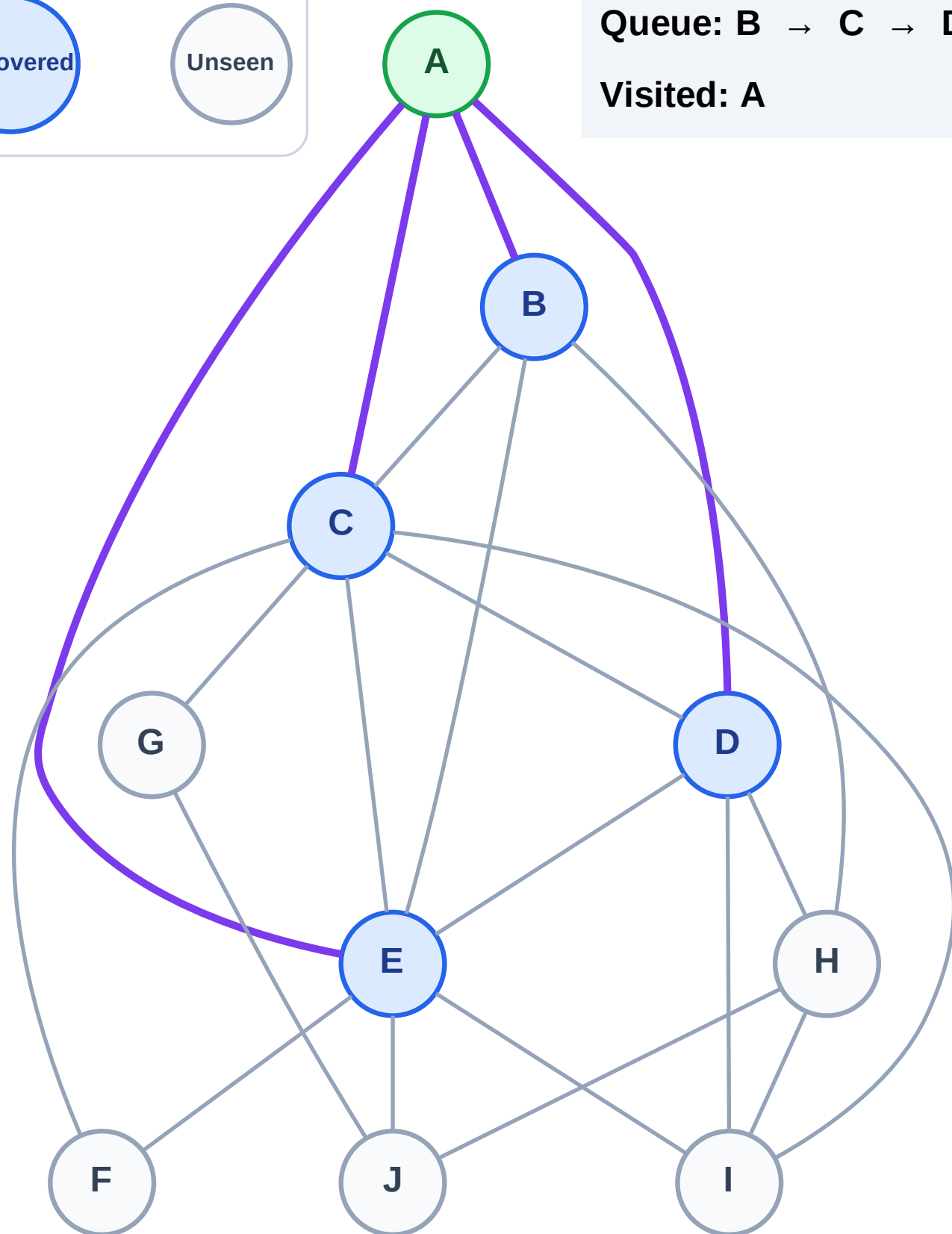
Step 3: Discover B, C, D, E from A

Legend



Queue: B → C → D → E

Visited: A



BFS Traversal

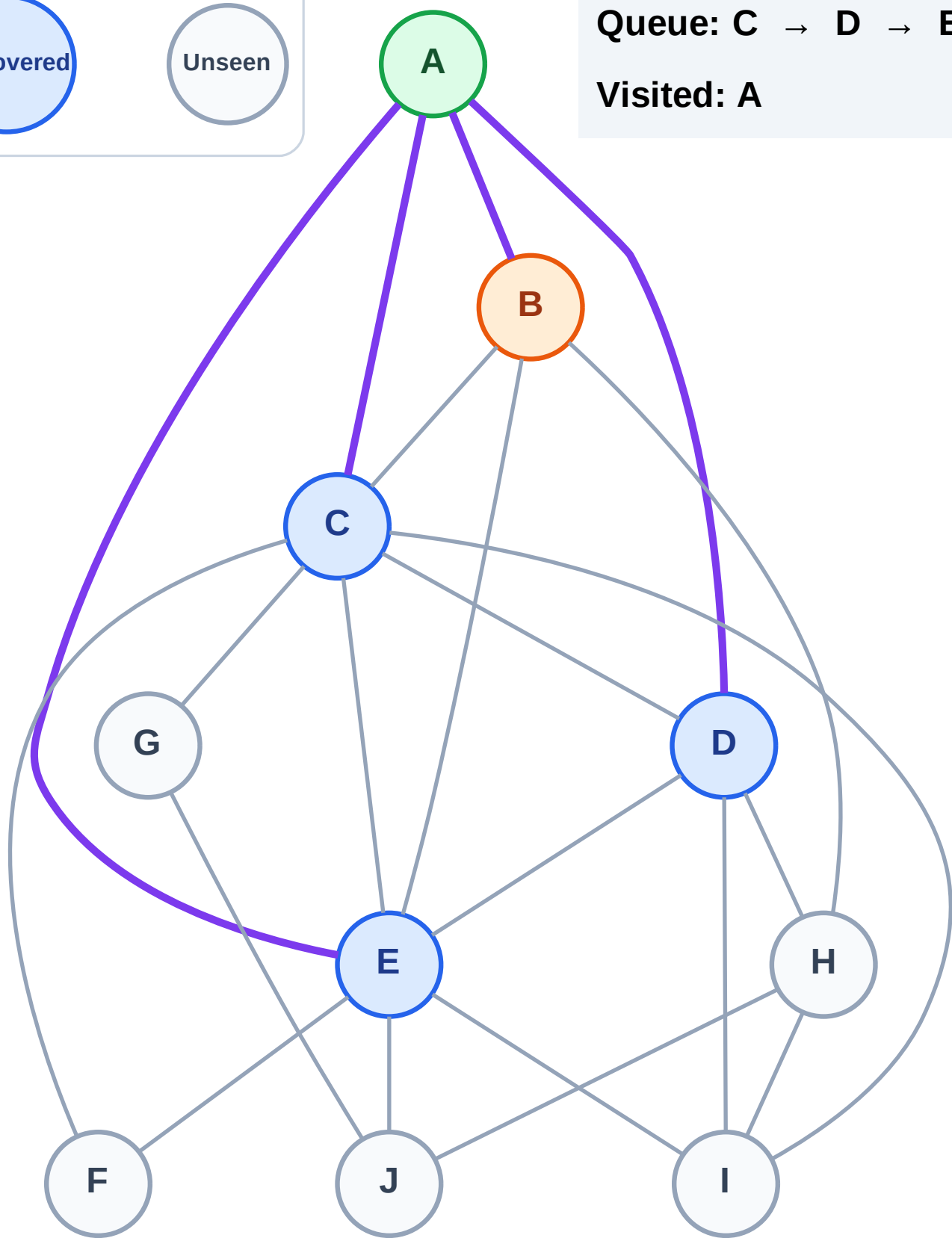
Step 4: Process node B

Legend



Queue: C → D → E

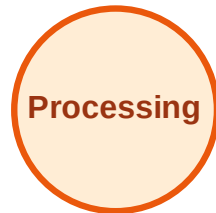
Visited: A



BFS Traversal

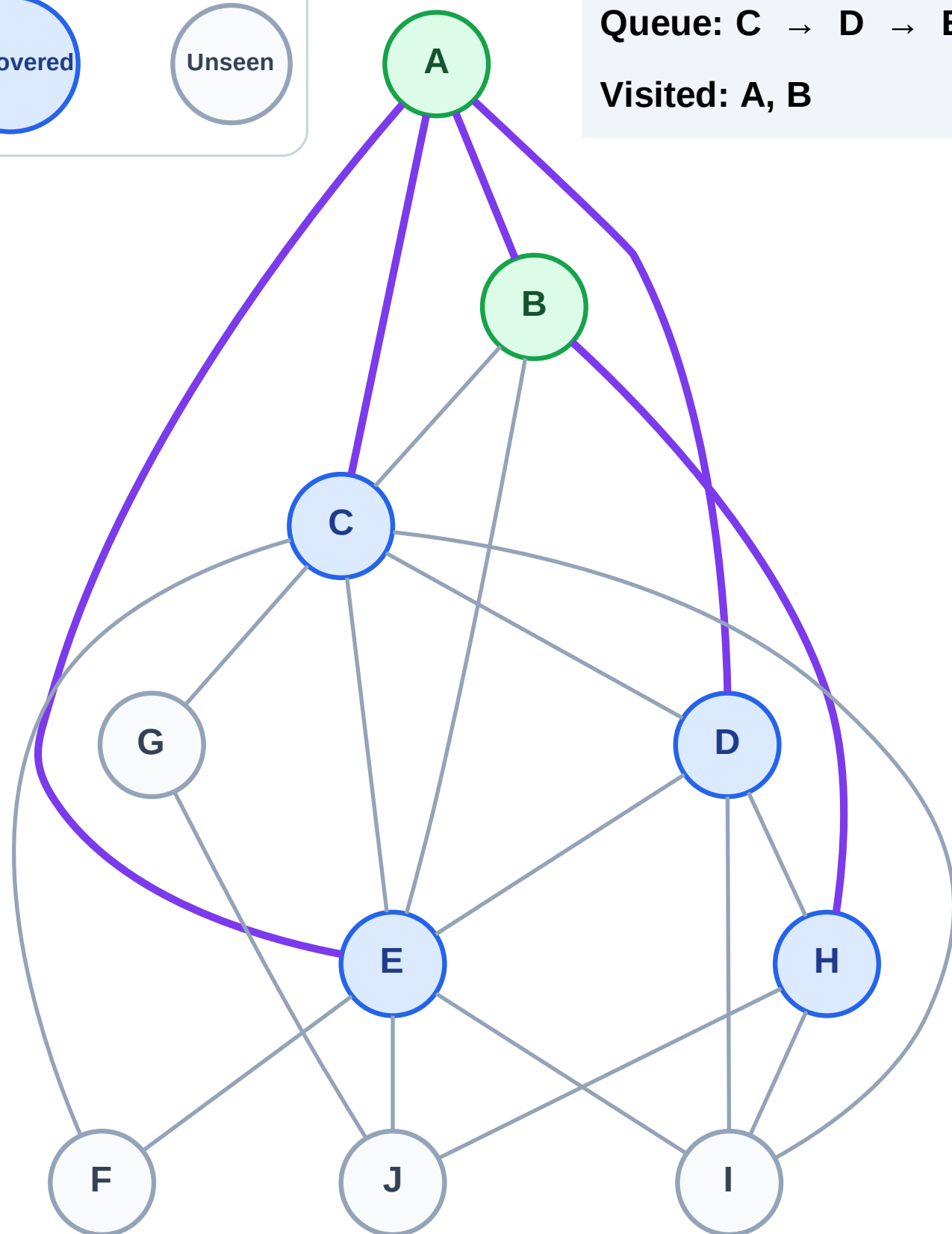
Step 5: Discover H from B

Legend



Queue: C → D → E → H

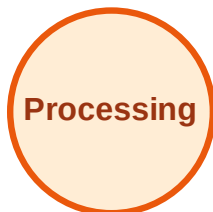
Visited: A, B



BFS Traversal

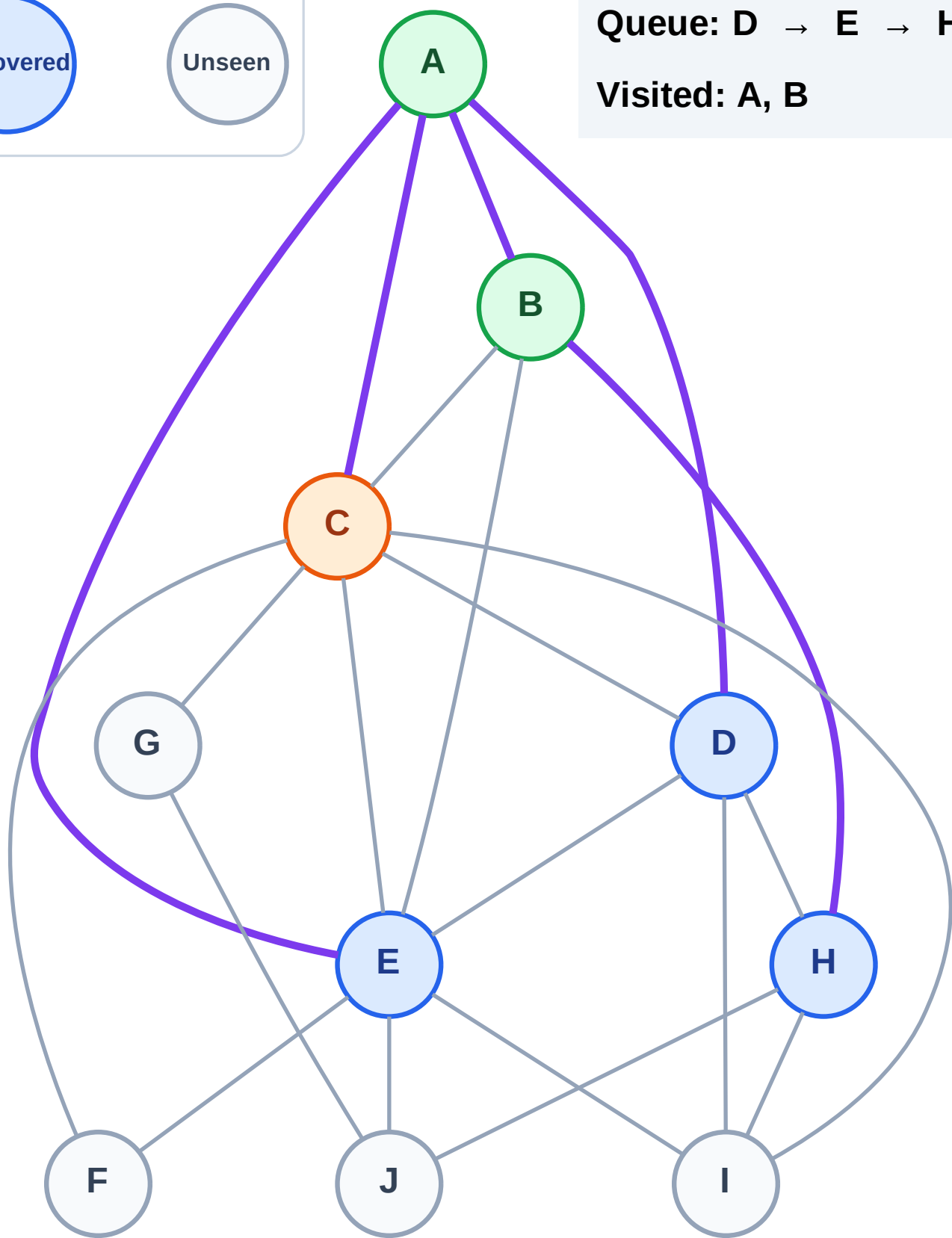
Step 6: Process node C

Legend



Queue: D → E → H

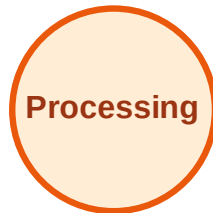
Visited: A, B



BFS Traversal

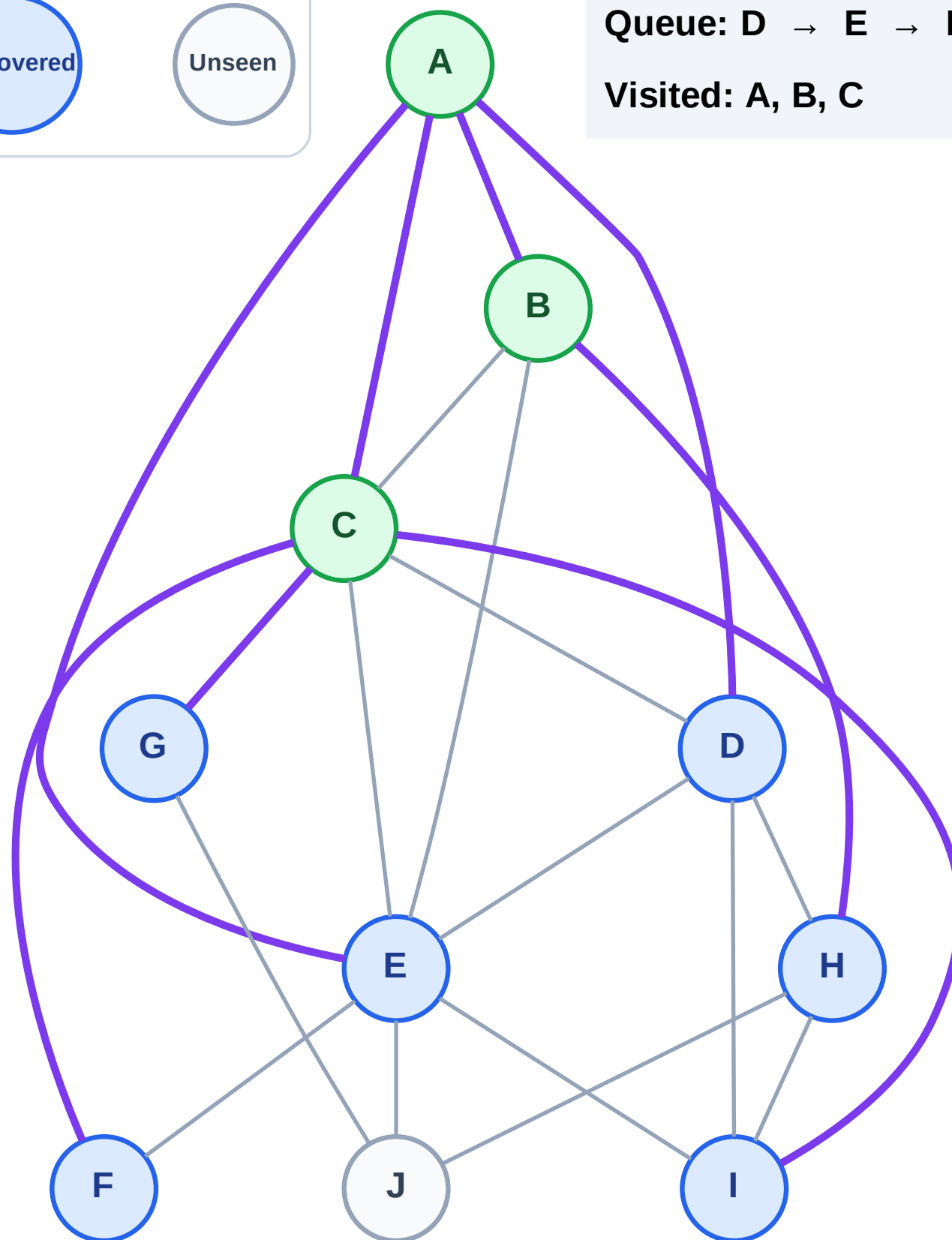
Step 7: Discover F, G, I from C

Legend



Queue: D → E → H → F → G → I

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

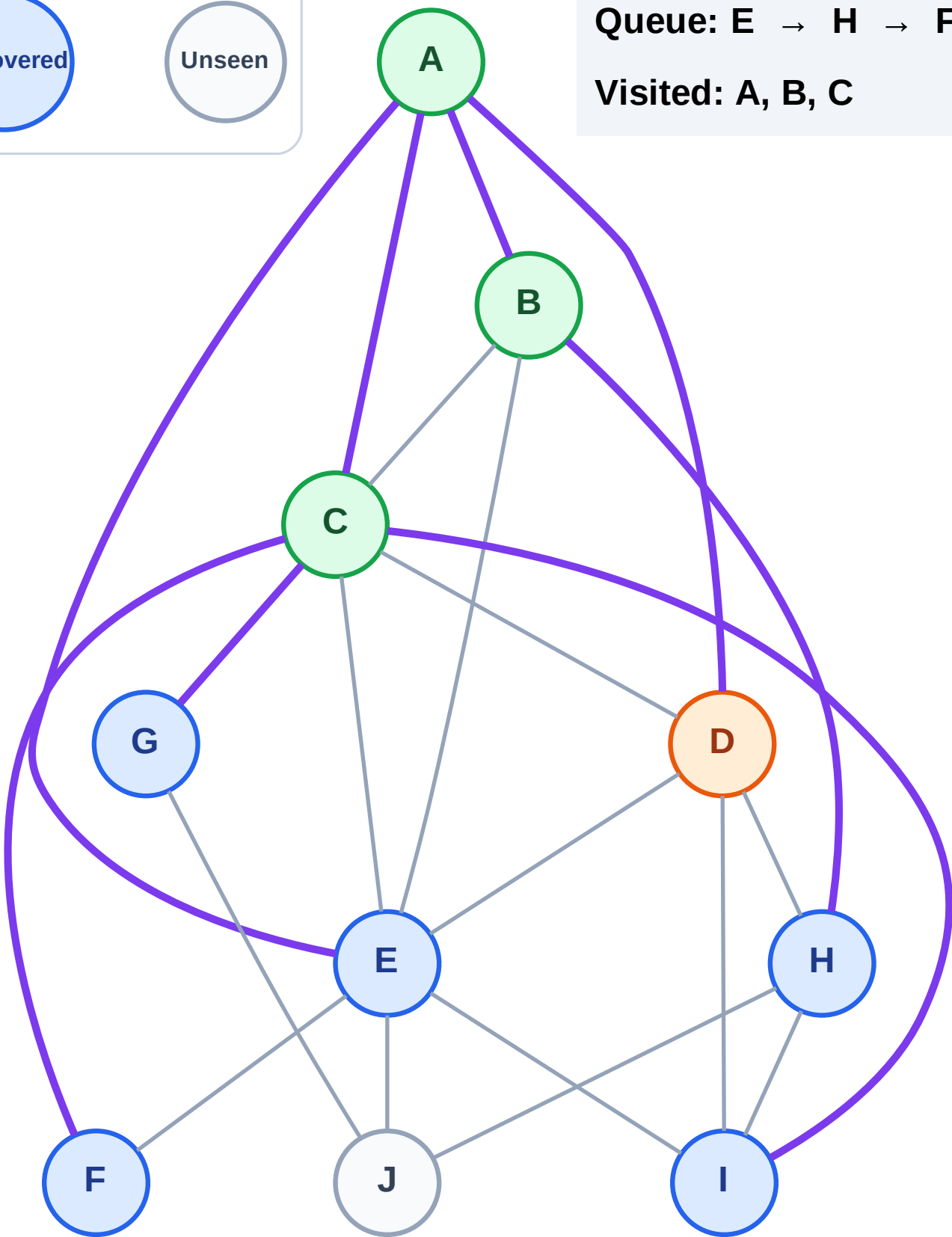
Complete

Processing

Discovered

Unseen

Queue: E → H → F → G → I
Visited: A, B, C



BFS Traversal

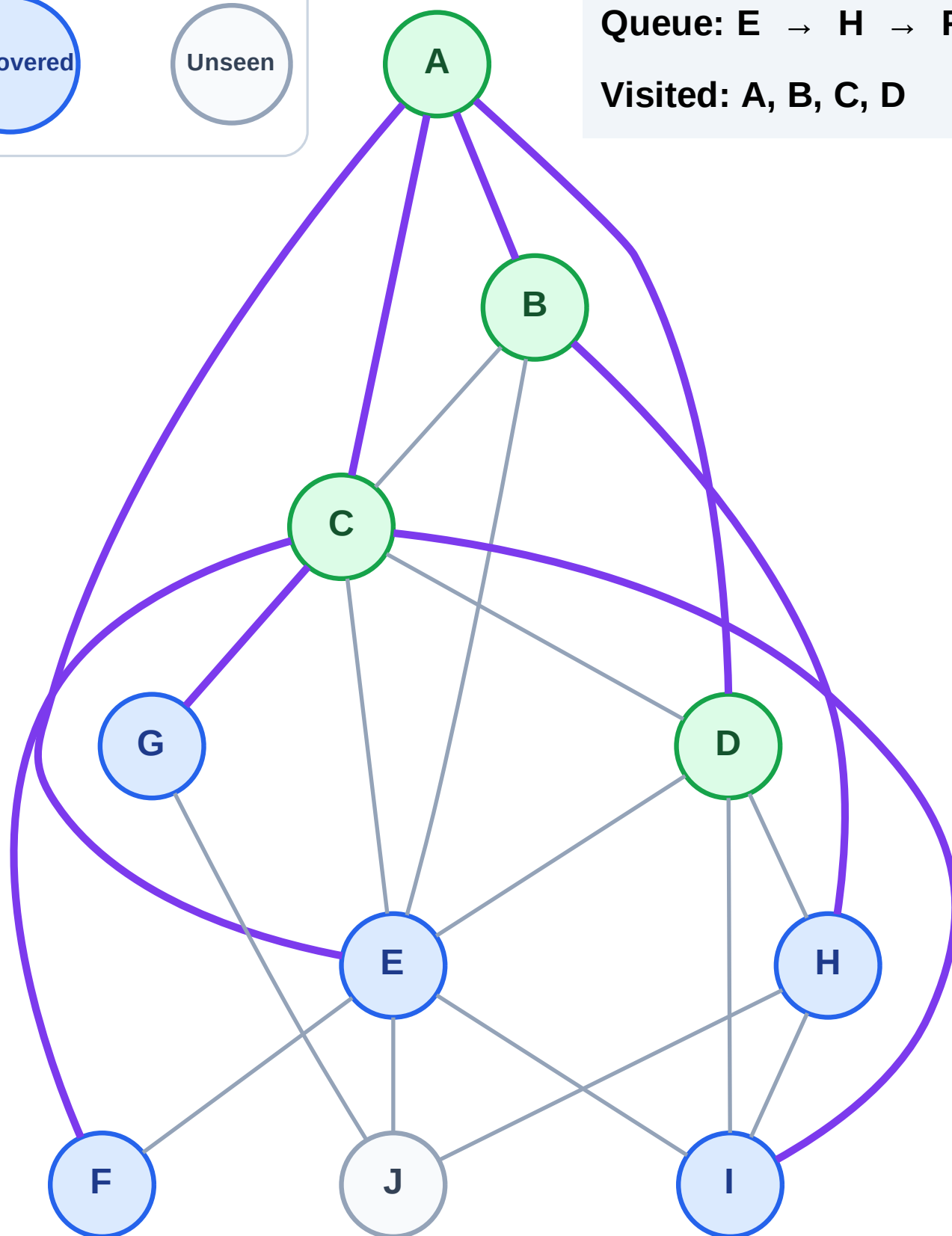
Step 9: No new neighbors from D

Legend



Queue: E → H → F → G → I

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

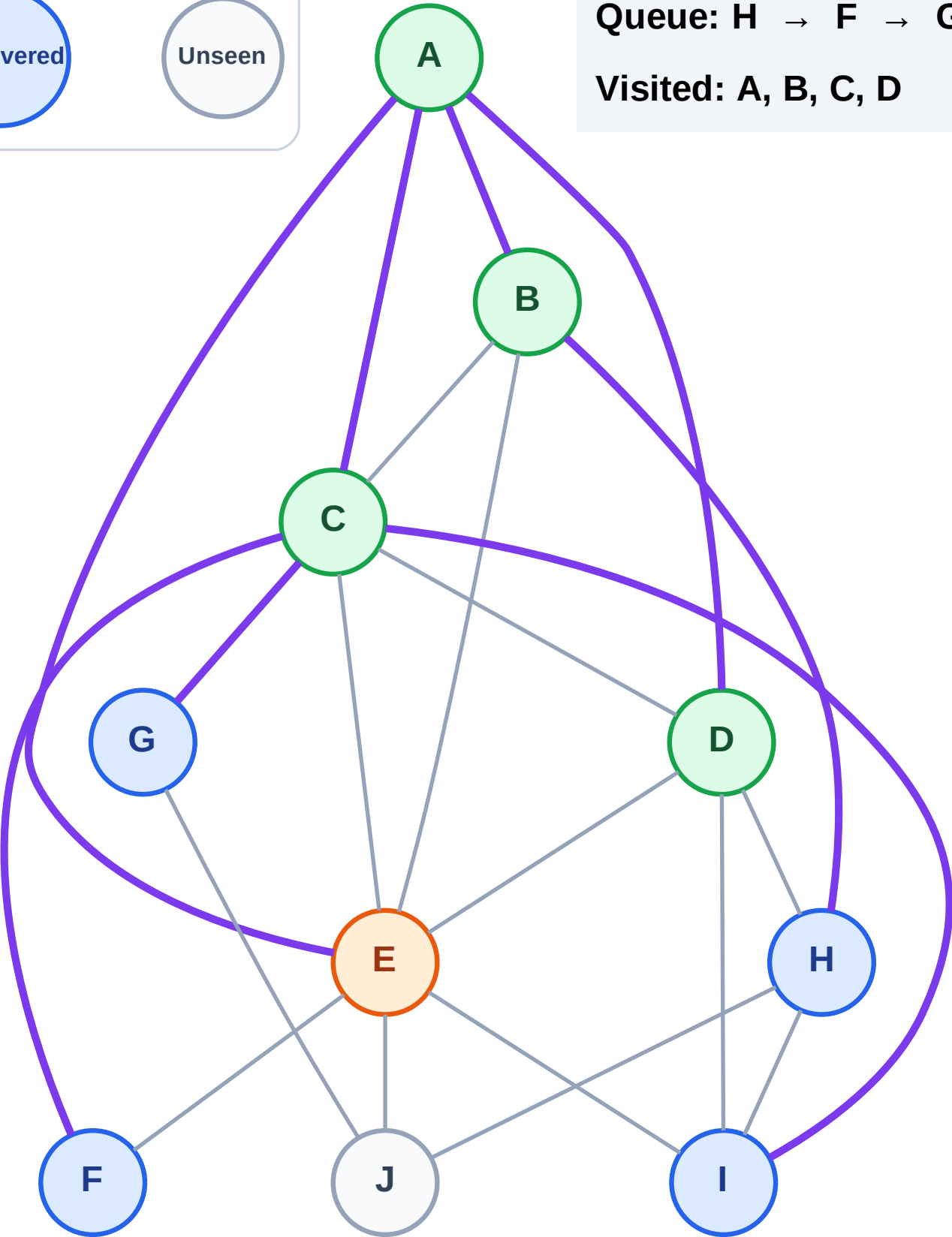
Processing

Discovered

Unseen

Queue: H → F → G → I

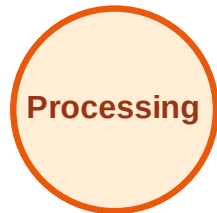
Visited: A, B, C, D



BFS Traversal

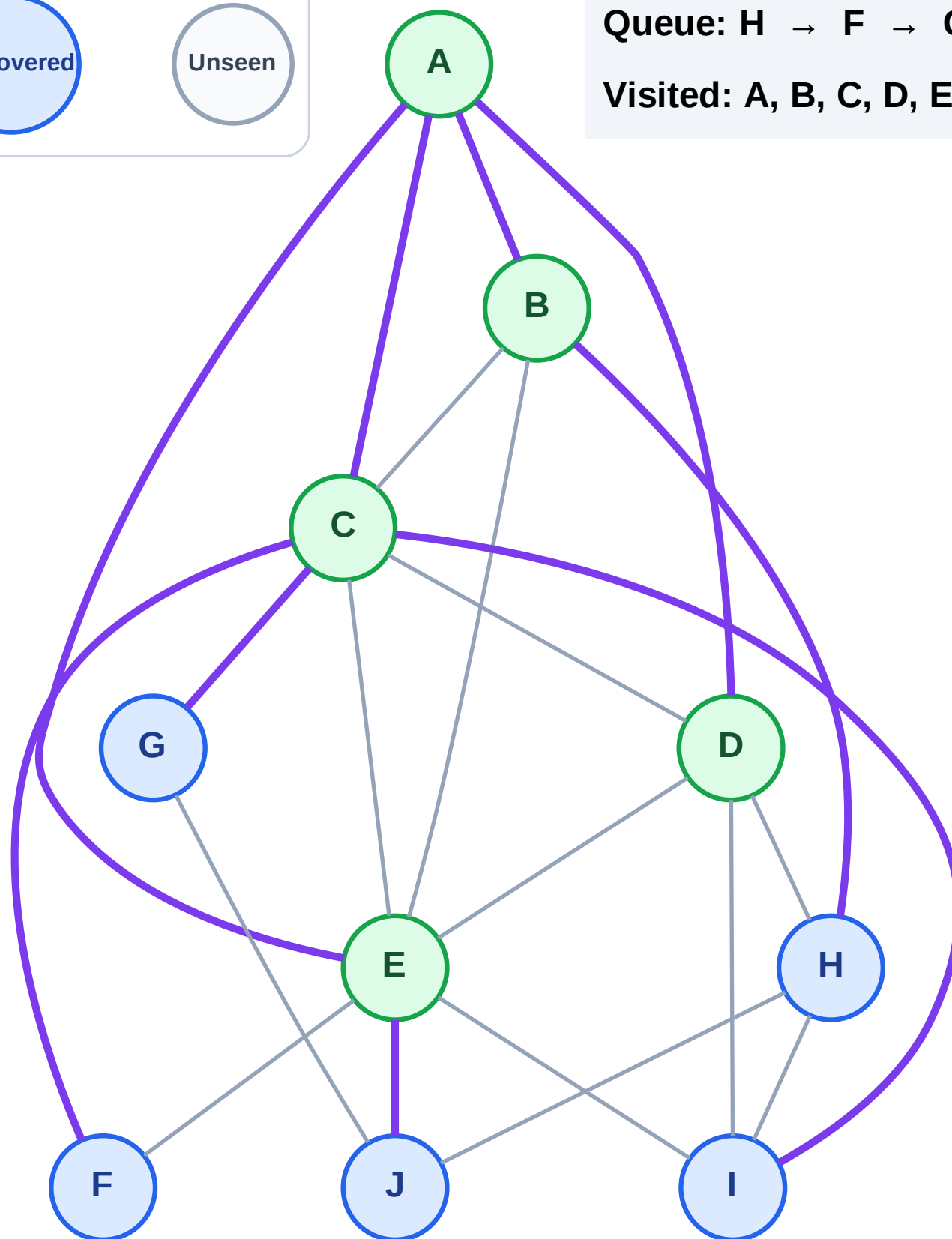
Step 11: Discover J from E

Legend



Queue: H → F → G → I → J

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node H

Legend

Complete

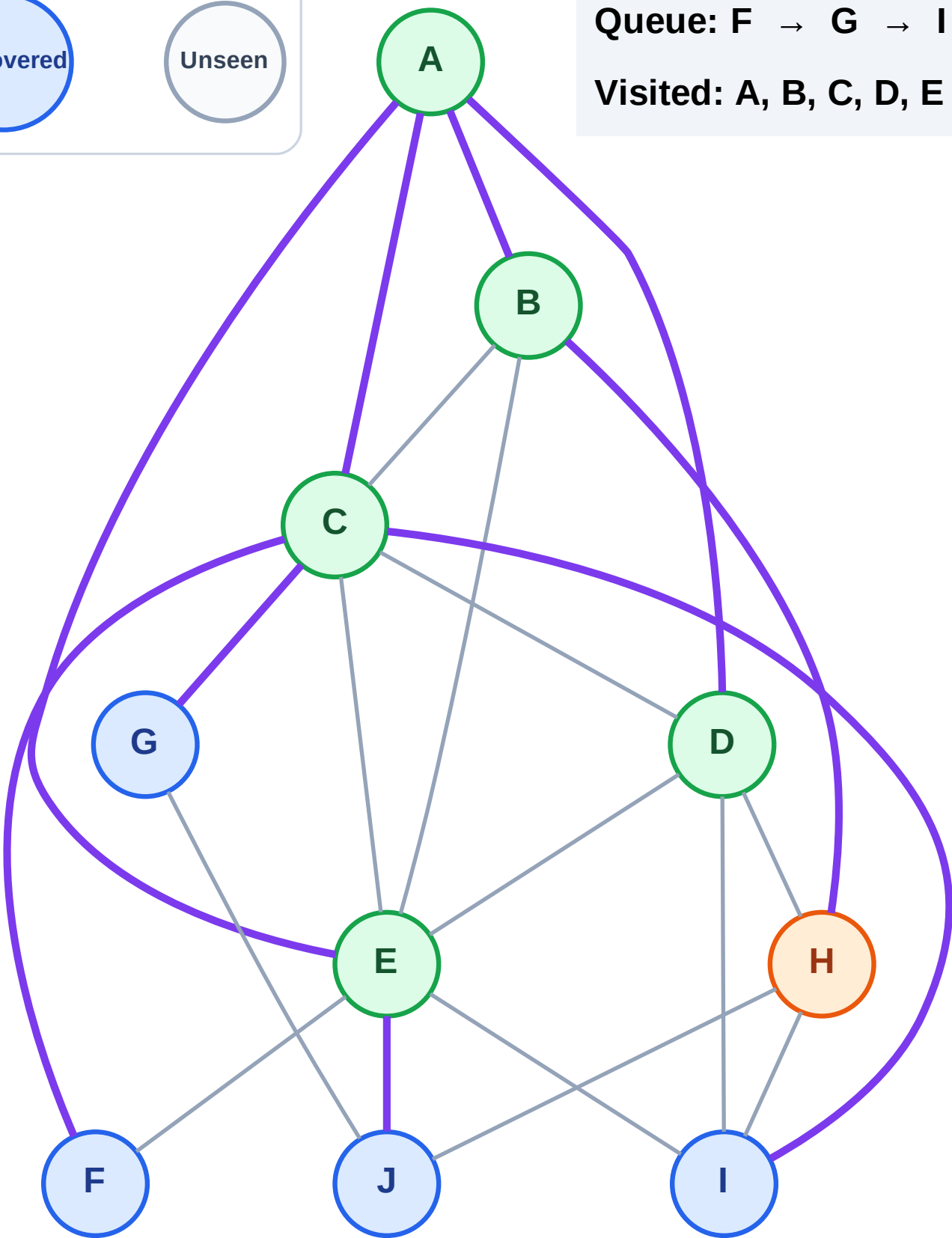
Processing

Discovered

Unseen

Queue: F → G → I → J

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from H

Legend

Complete

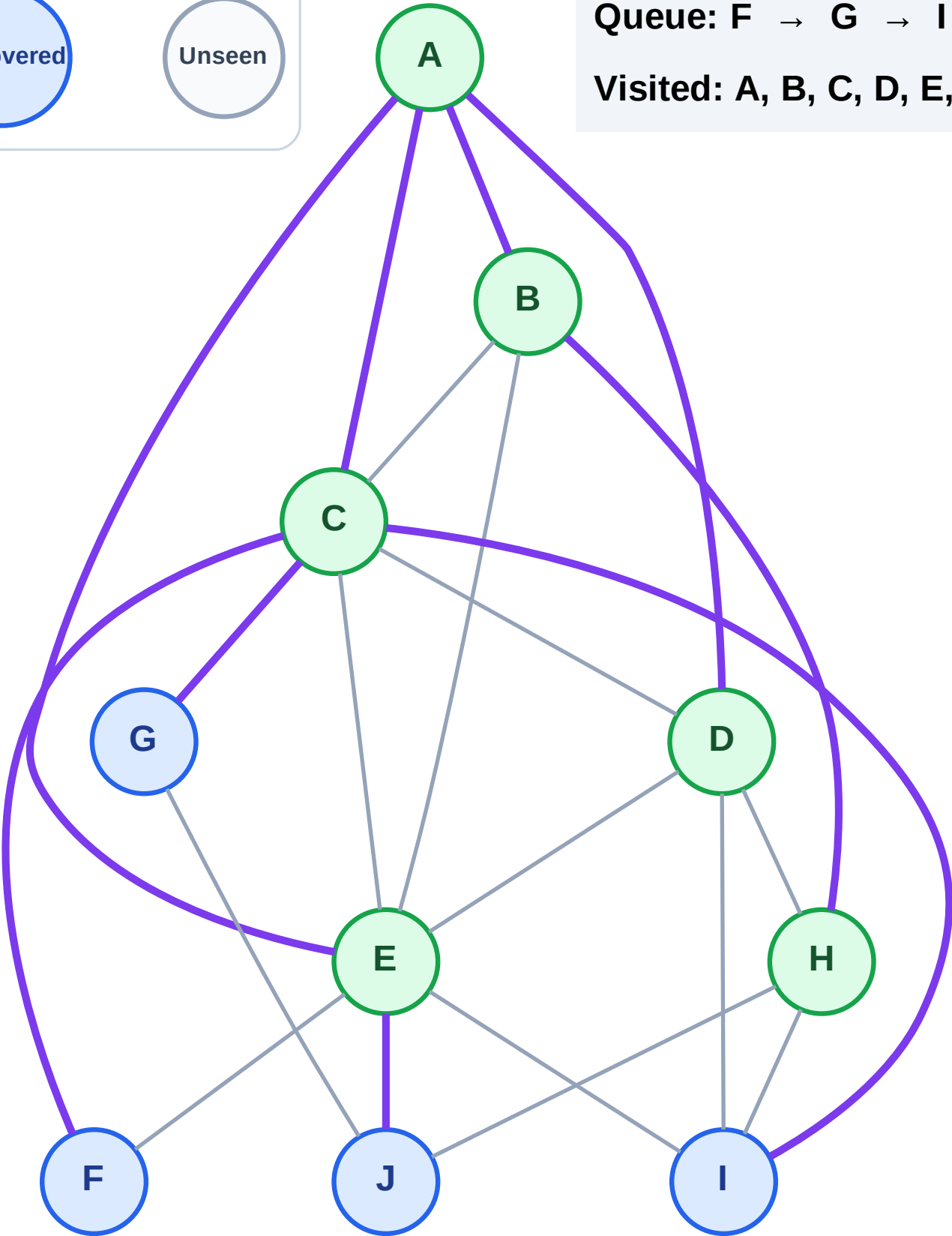
Processing

Discovered

Unseen

Queue: F → G → I → J

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node F

Legend

Complete

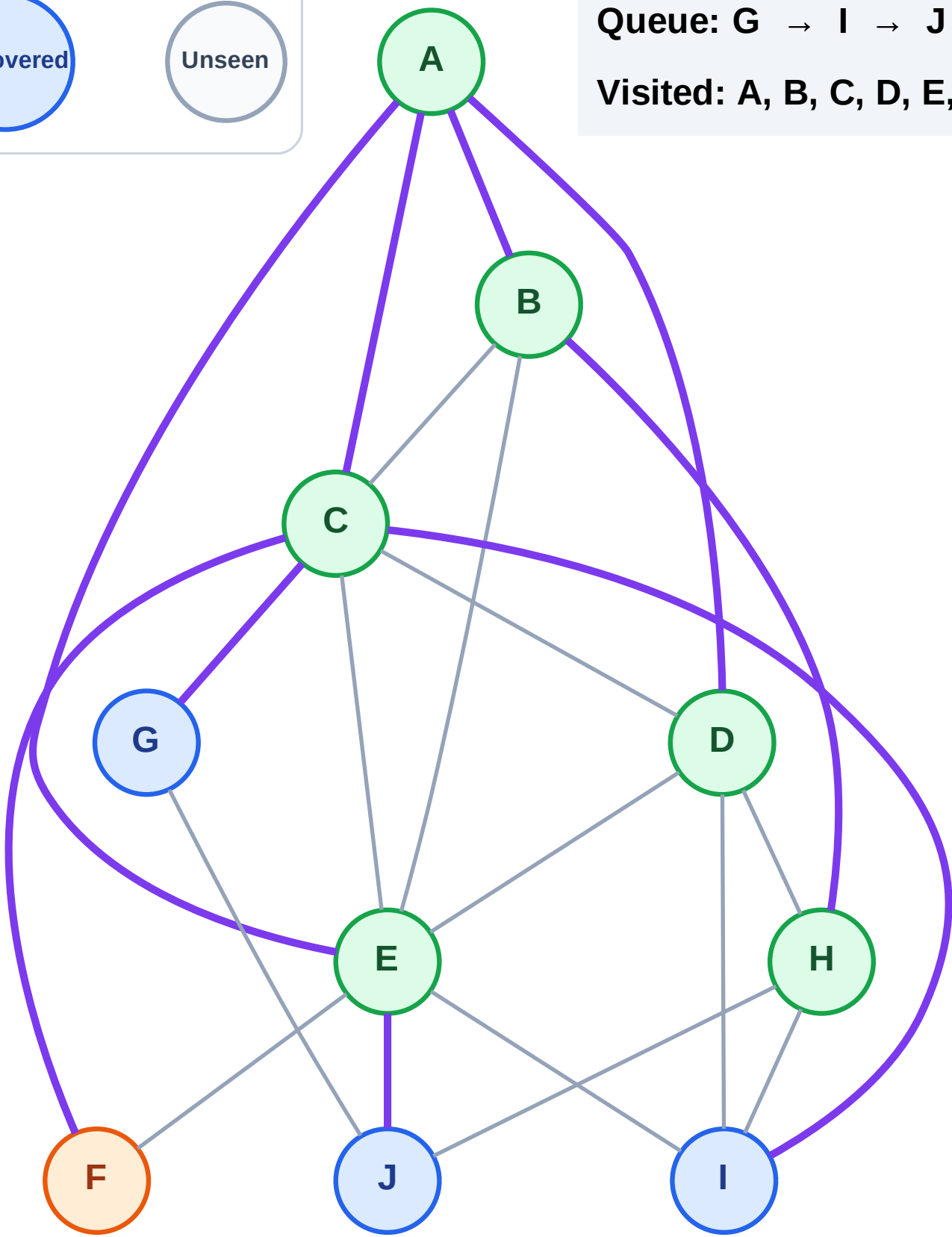
Processing

Discovered

Unseen

Queue: G → I → J

Visited: A, B, C, D, E, H



BFS Traversal

Step 15: No new neighbors from F

Legend

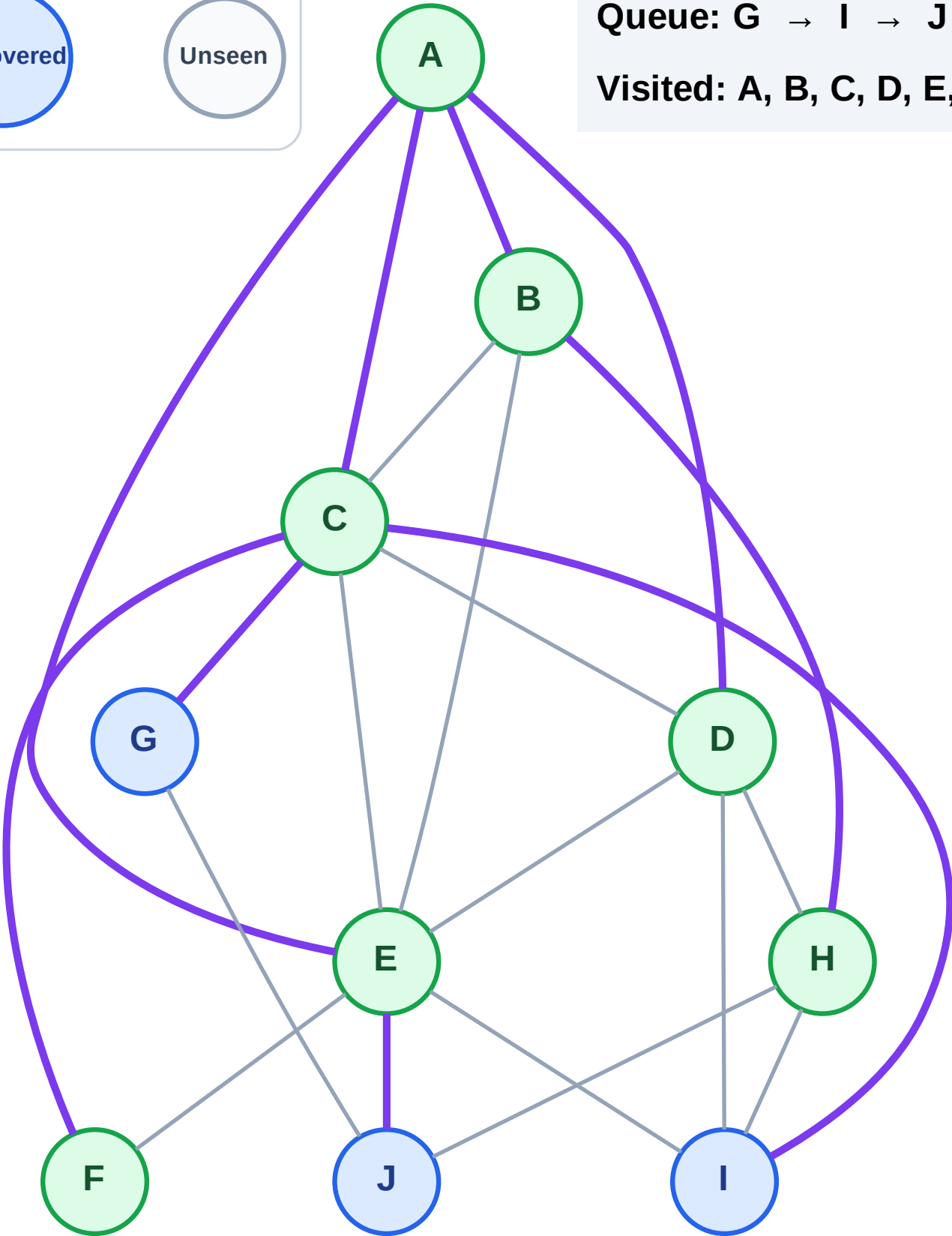
Complete

Processing

Discovered

Unseen

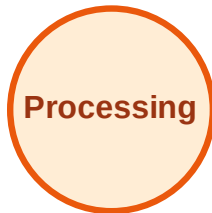
Queue: G → I → J
Visited: A, B, C, D, E, F, H



BFS Traversal

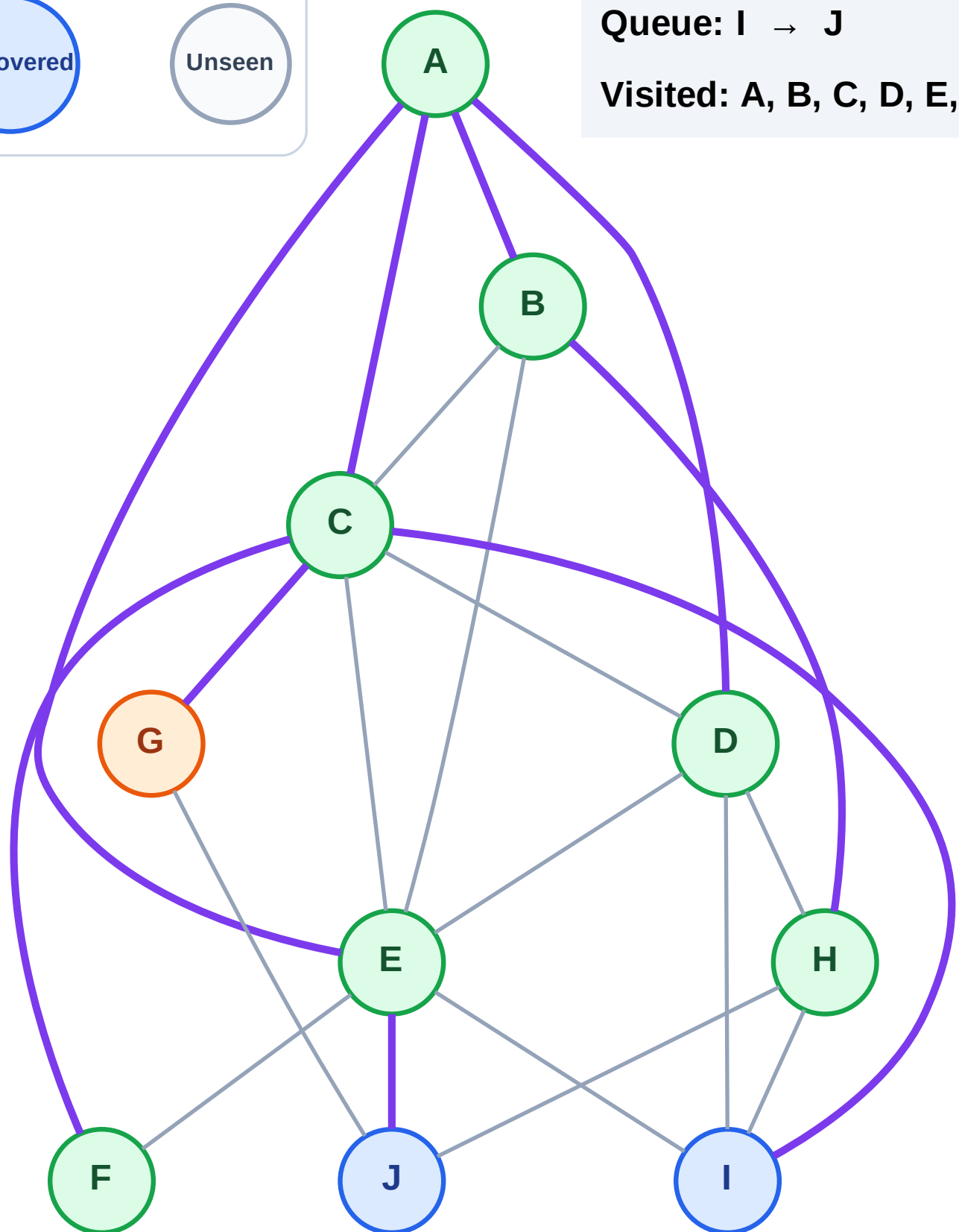
Step 16: Process node G

Legend



Queue: I → J

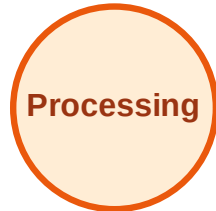
Visited: A, B, C, D, E, F, H



BFS Traversal

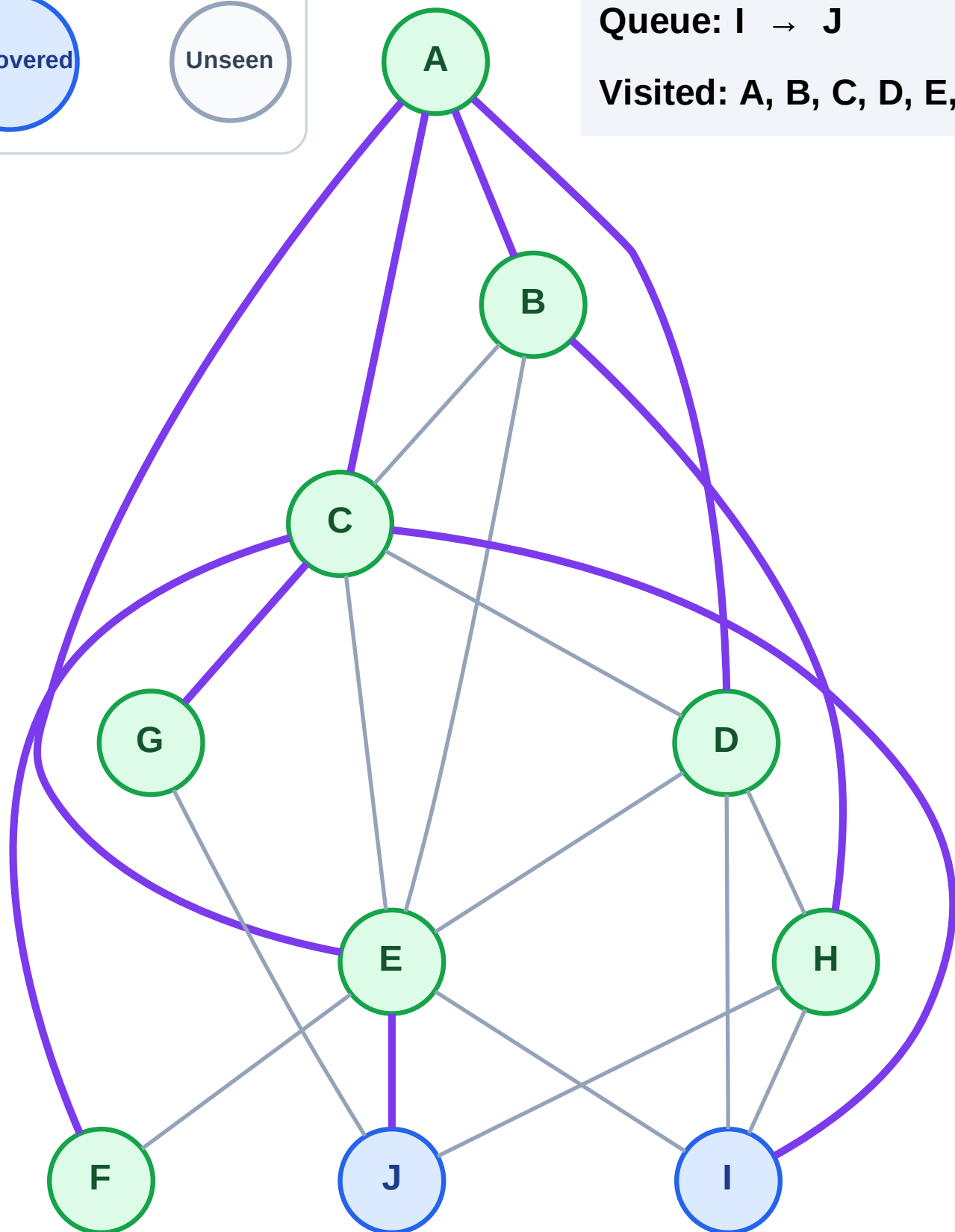
Step 17: No new neighbors from G

Legend



Queue: I → J

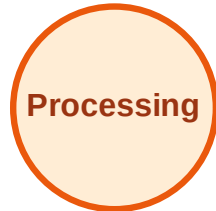
Visited: A, B, C, D, E, F, G, H



BFS Traversal

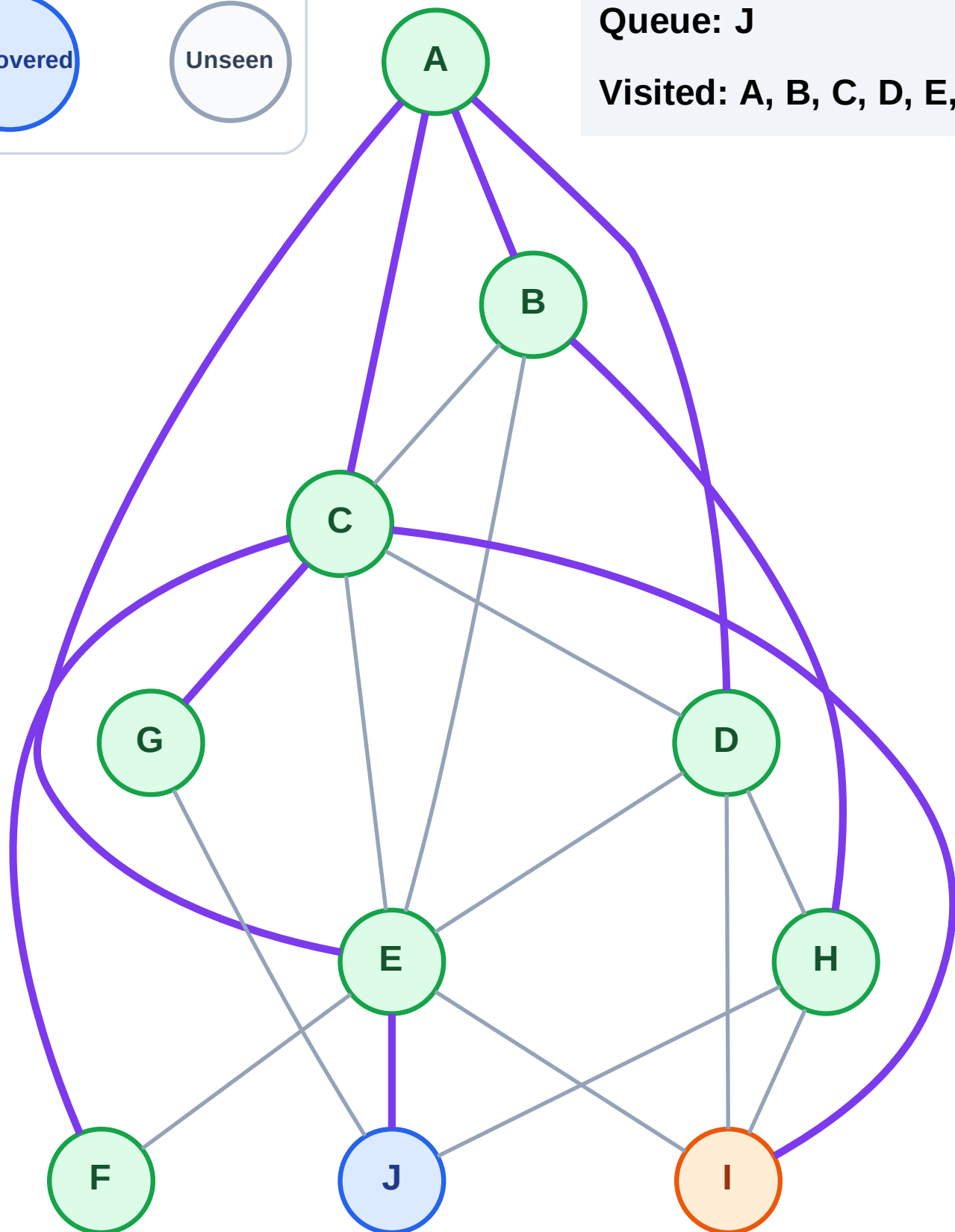
Step 18: Process node I

Legend



Queue: J

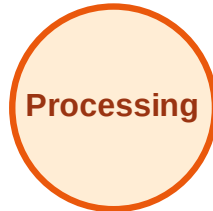
Visited: A, B, C, D, E, F, G, H



BFS Traversal

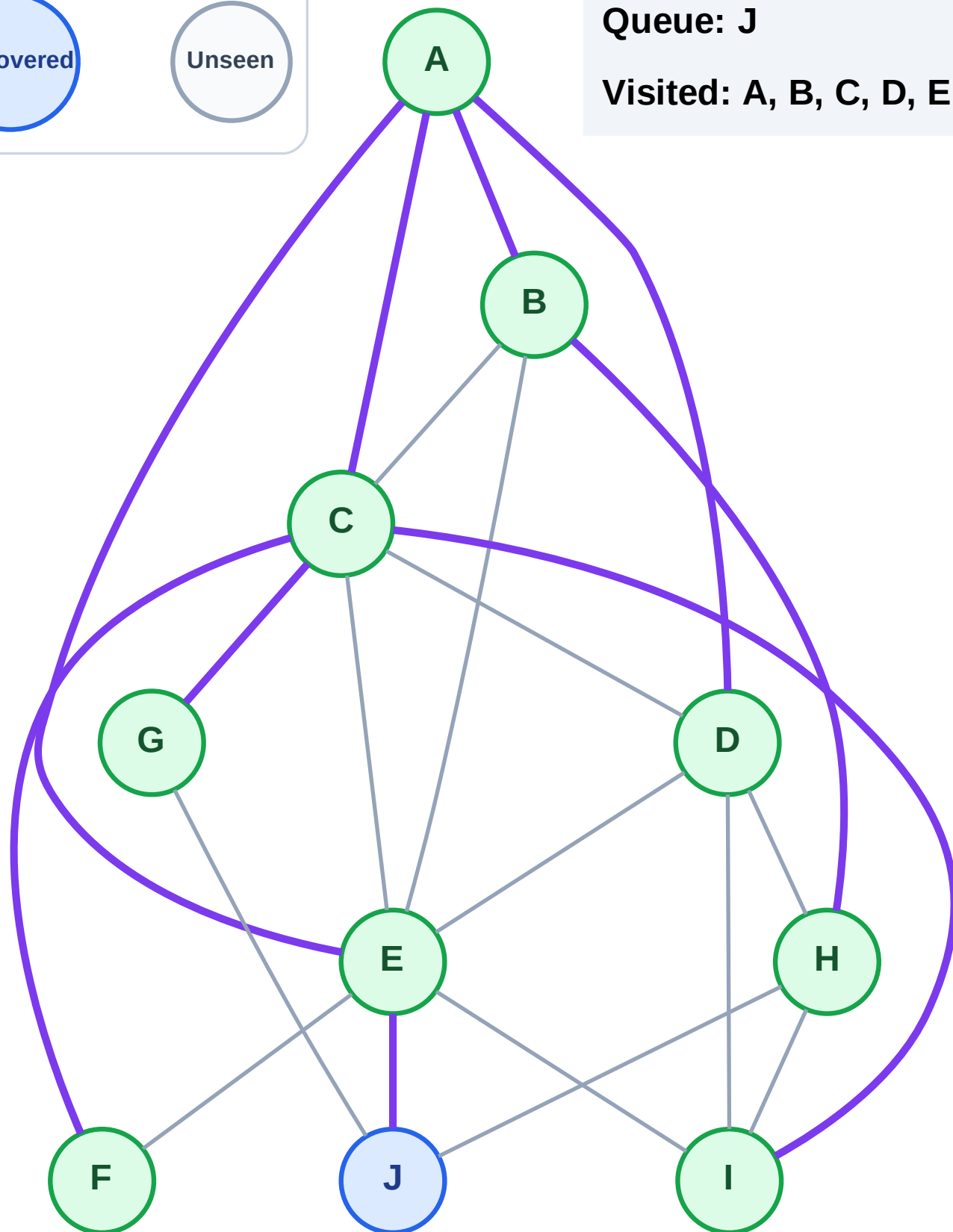
Step 19: No new neighbors from I

Legend



Queue: J

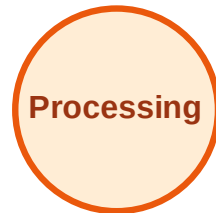
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

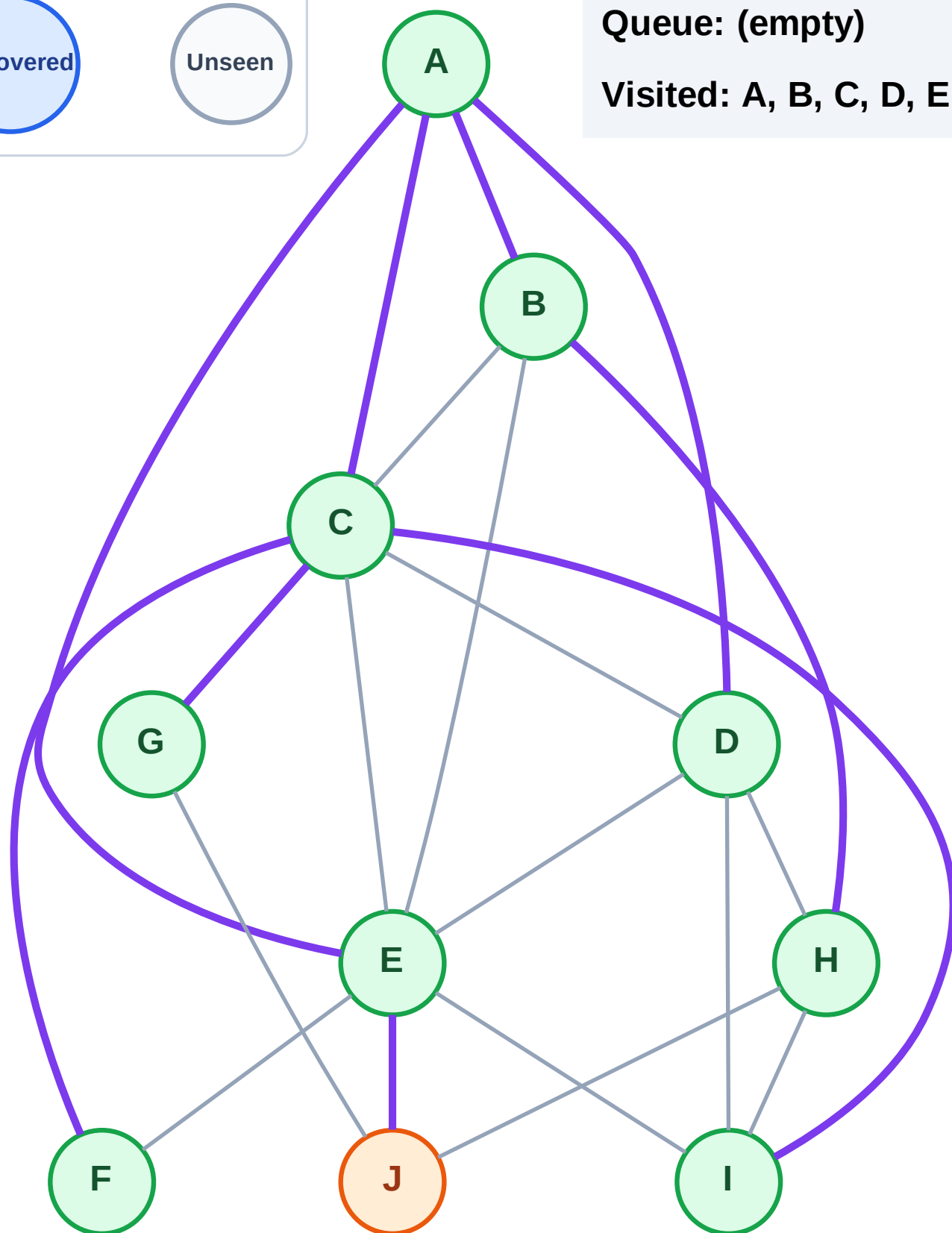
Step 20: Process node J

Legend



Queue: (empty)

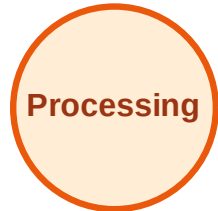
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

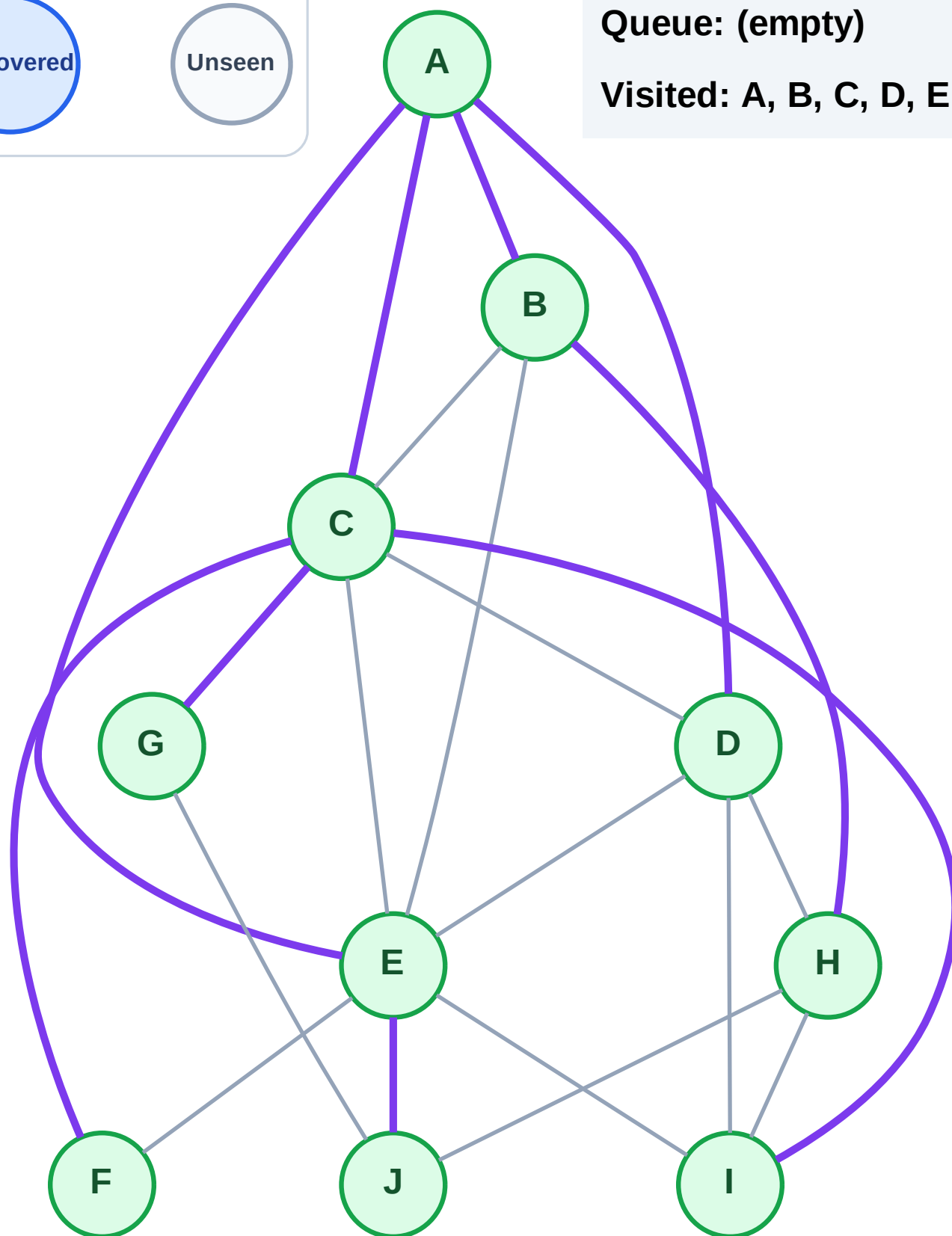
Step 21: No new neighbors from J

Legend



Queue: (empty)

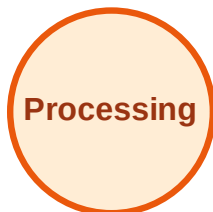
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

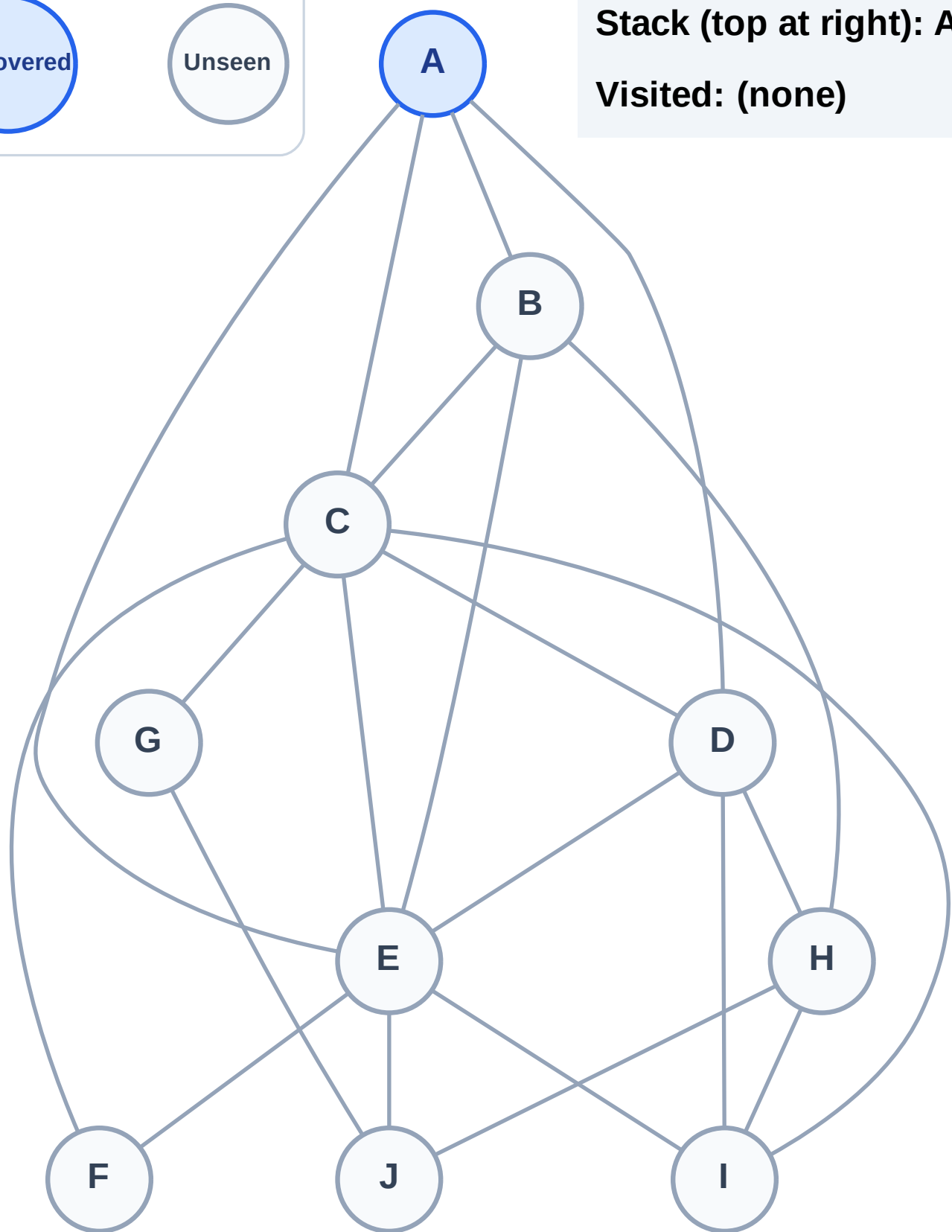
Step 1: Start at A

Legend



Stack (top at right): A

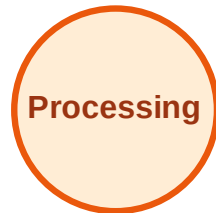
Visited: (none)



DFS Traversal

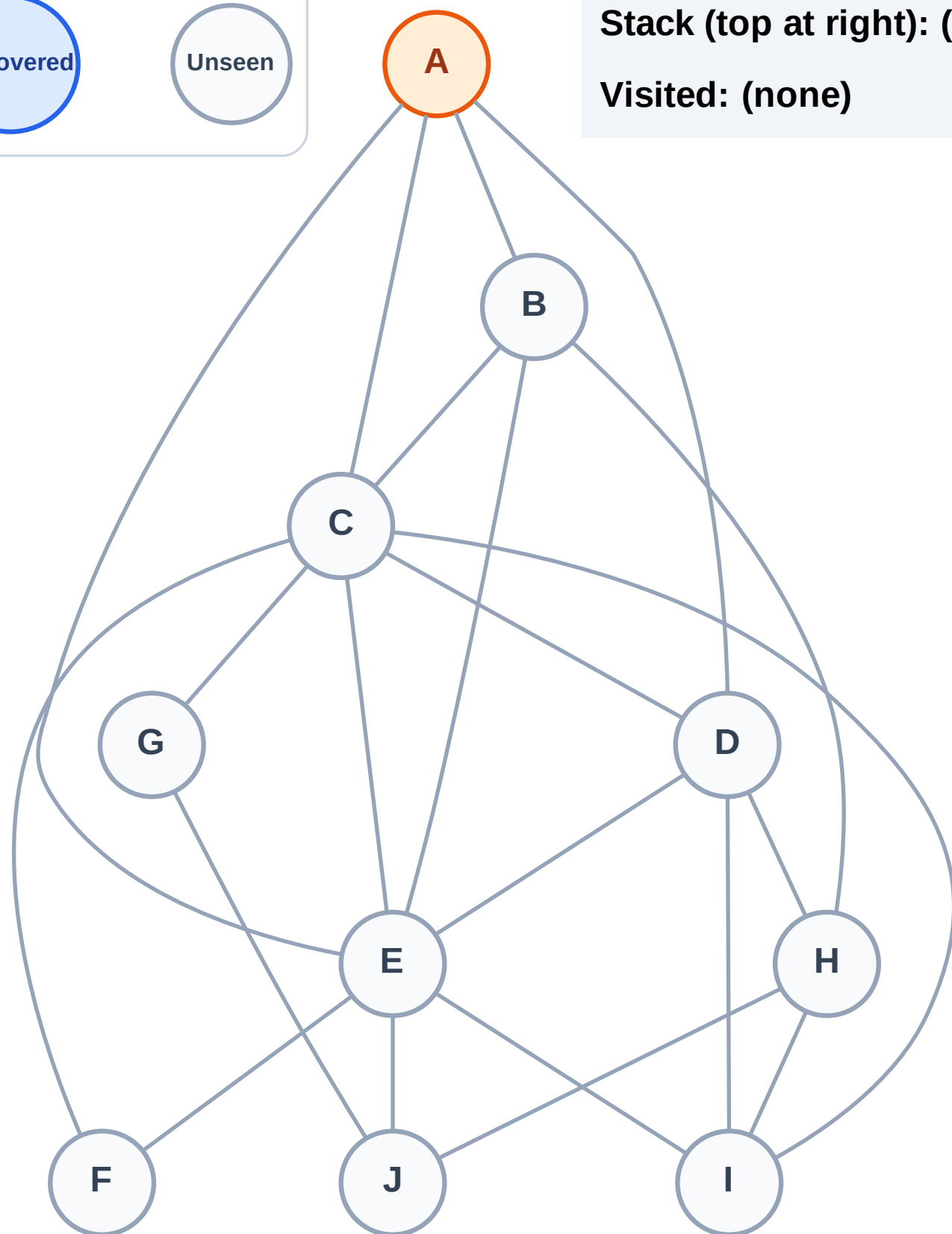
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover E, D, C, B from A

Legend

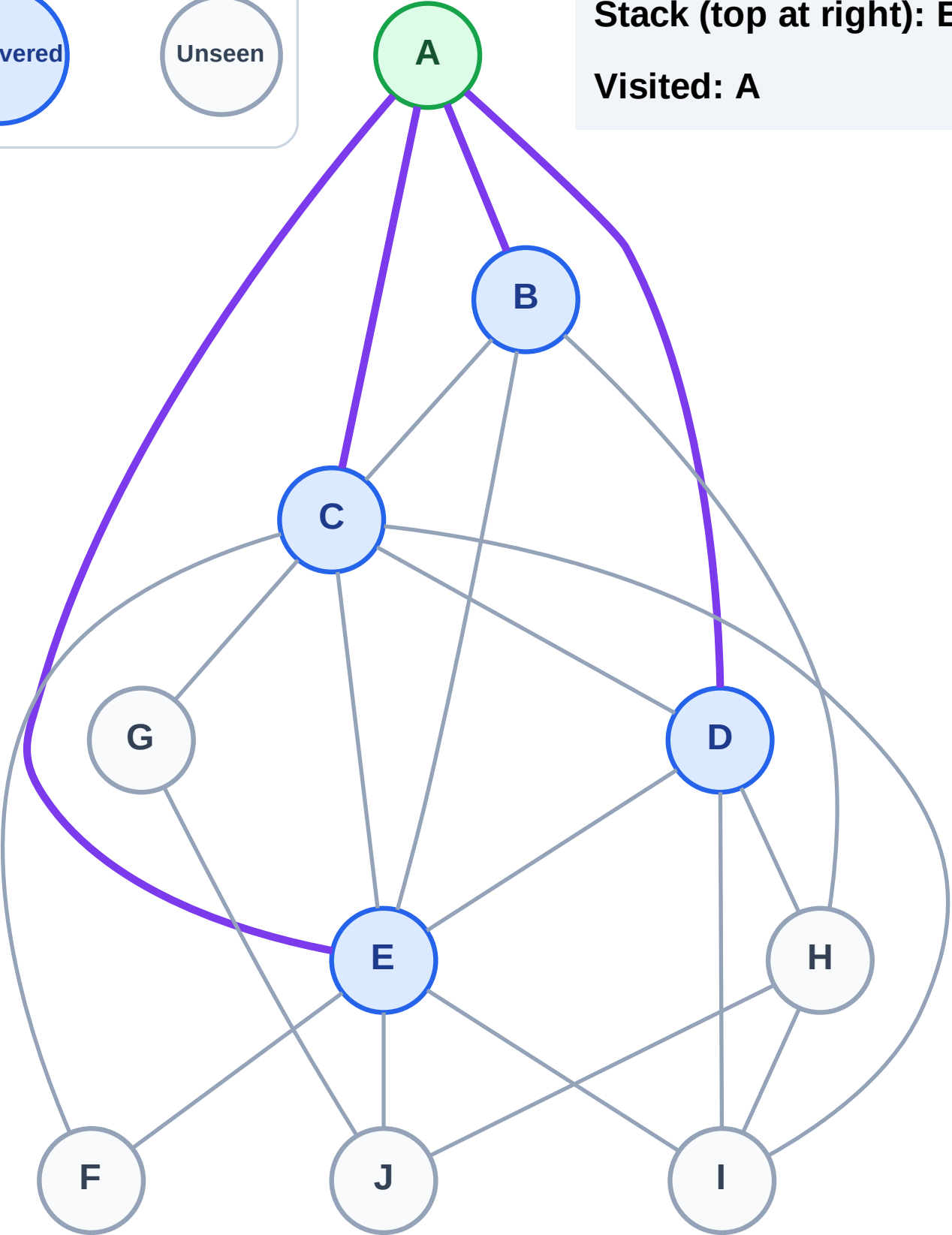
Complete

Processing

Discovered

Unseen

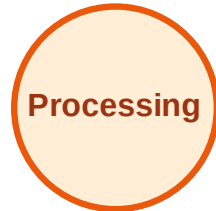
Stack (top at right): E | D | C | B
Visited: A



DFS Traversal

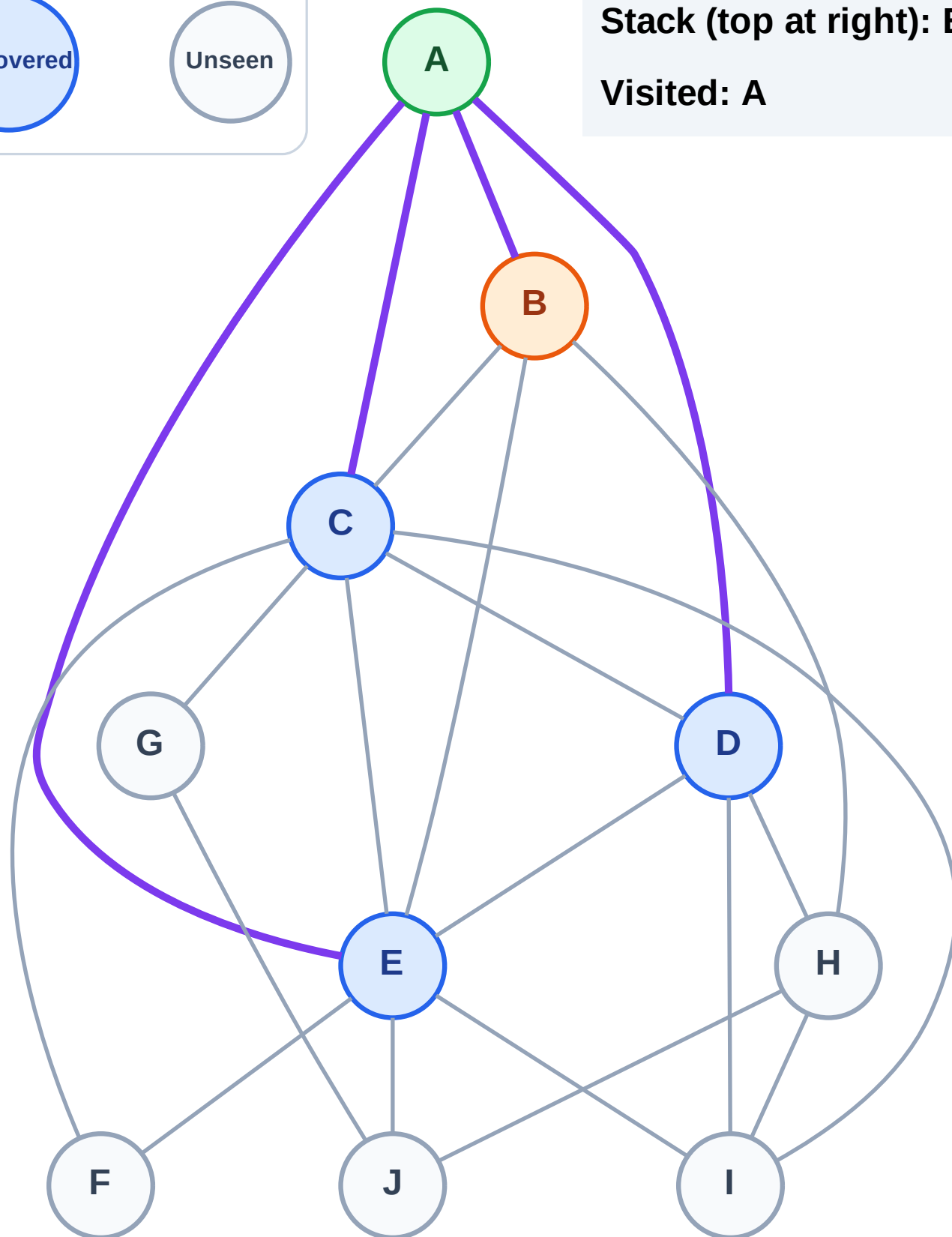
Step 4: Process node B

Legend



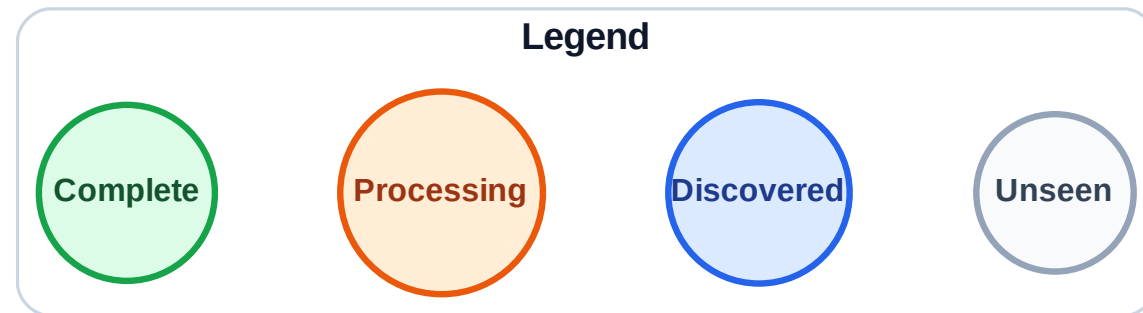
Stack (top at right): E | D | C

Visited: A

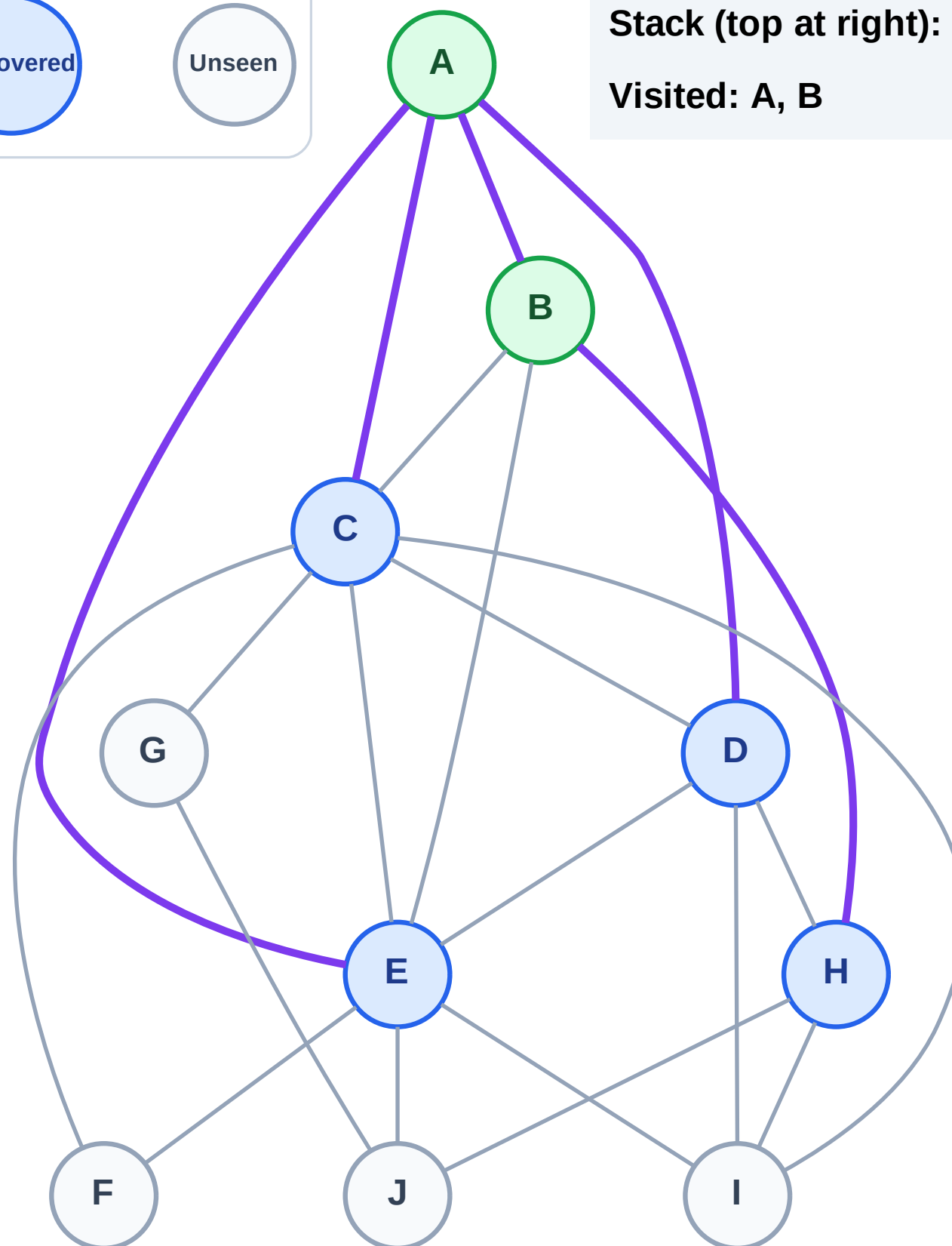


DFS Traversal

Step 5: Discover H from B



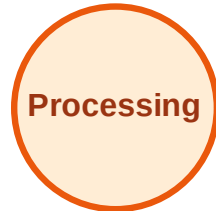
Stack (top at right): E | D | C | H
Visited: A, B



DFS Traversal

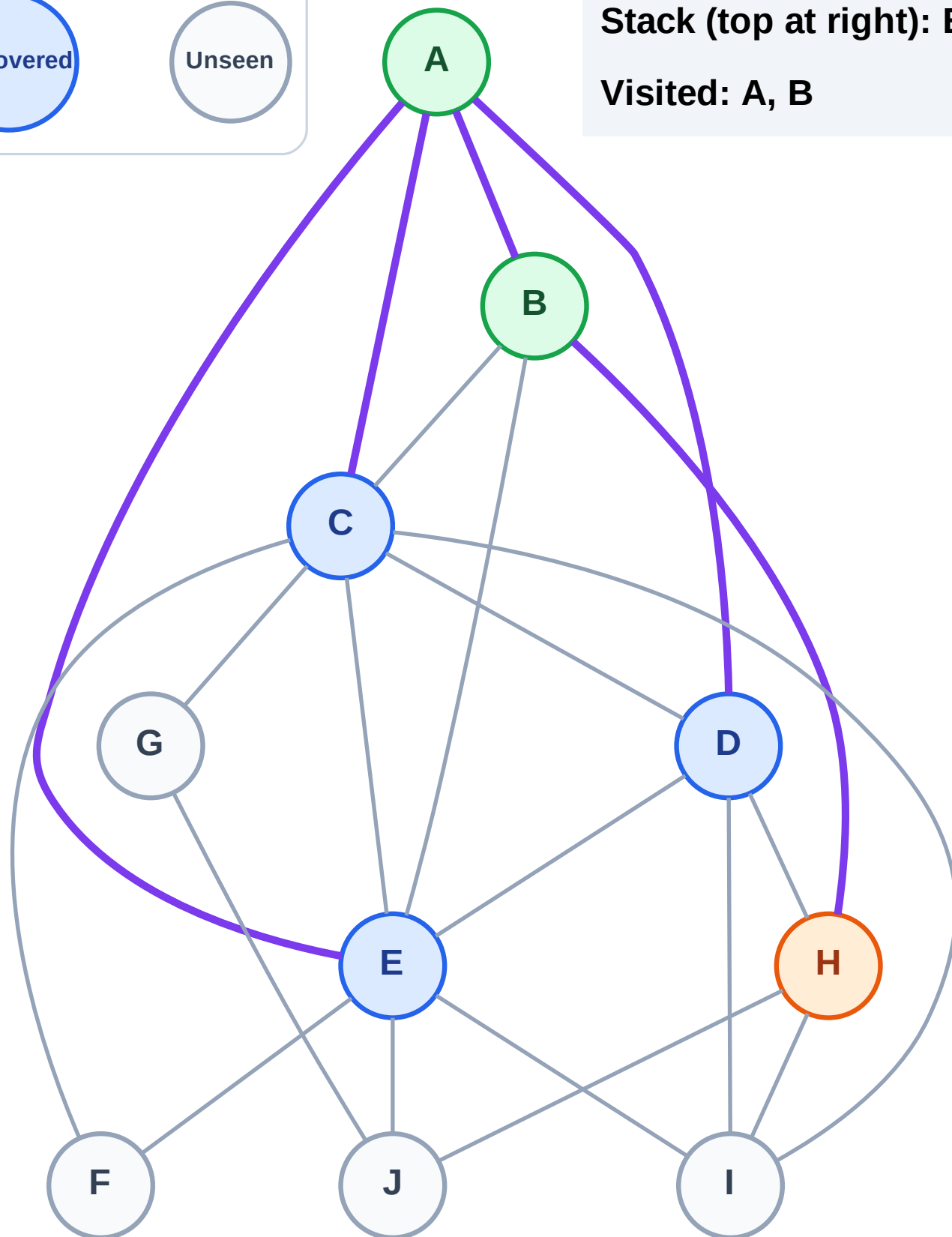
Step 6: Process node H

Legend



Stack (top at right): E | D | C

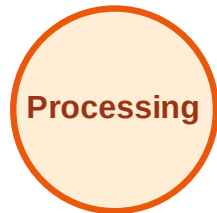
Visited: A, B



DFS Traversal

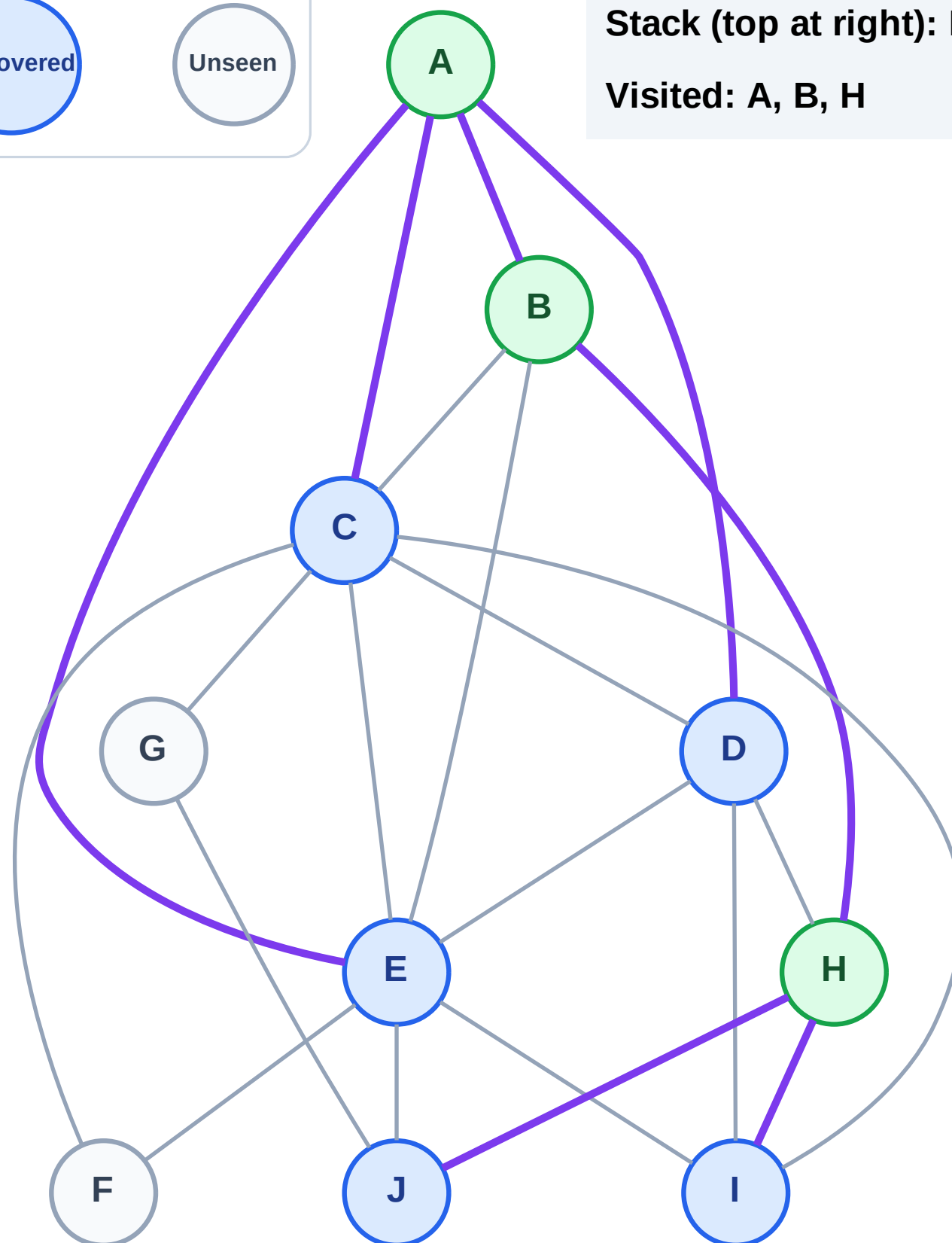
Step 7: Discover J, I from H

Legend



Stack (top at right): E | D | C | J | I

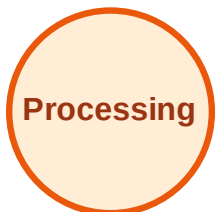
Visited: A, B, H



DFS Traversal

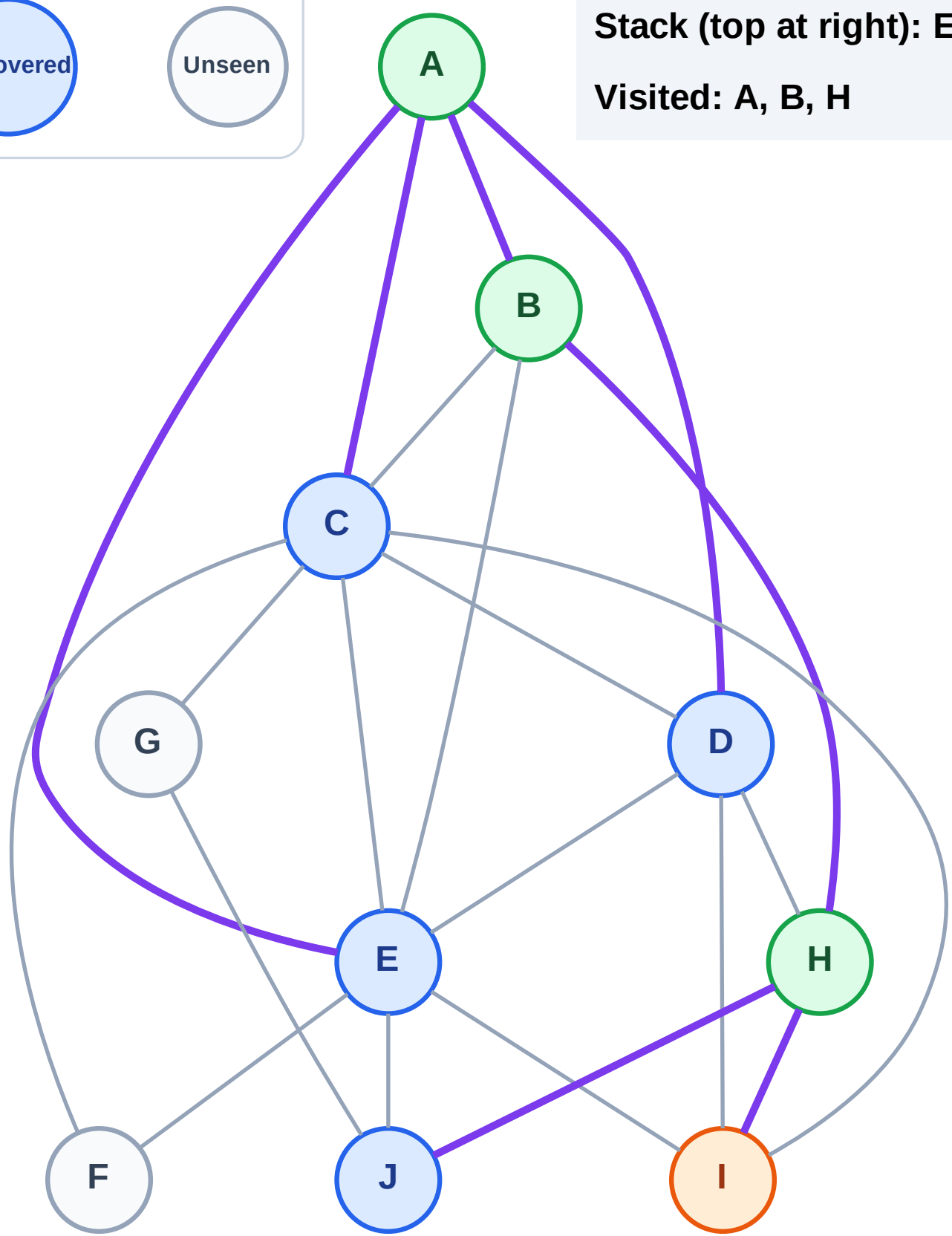
Step 8: Process node I

Legend



Stack (top at right): E | D | C | J

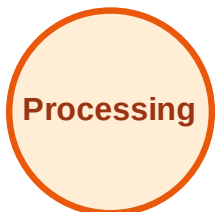
Visited: A, B, H



DFS Traversal

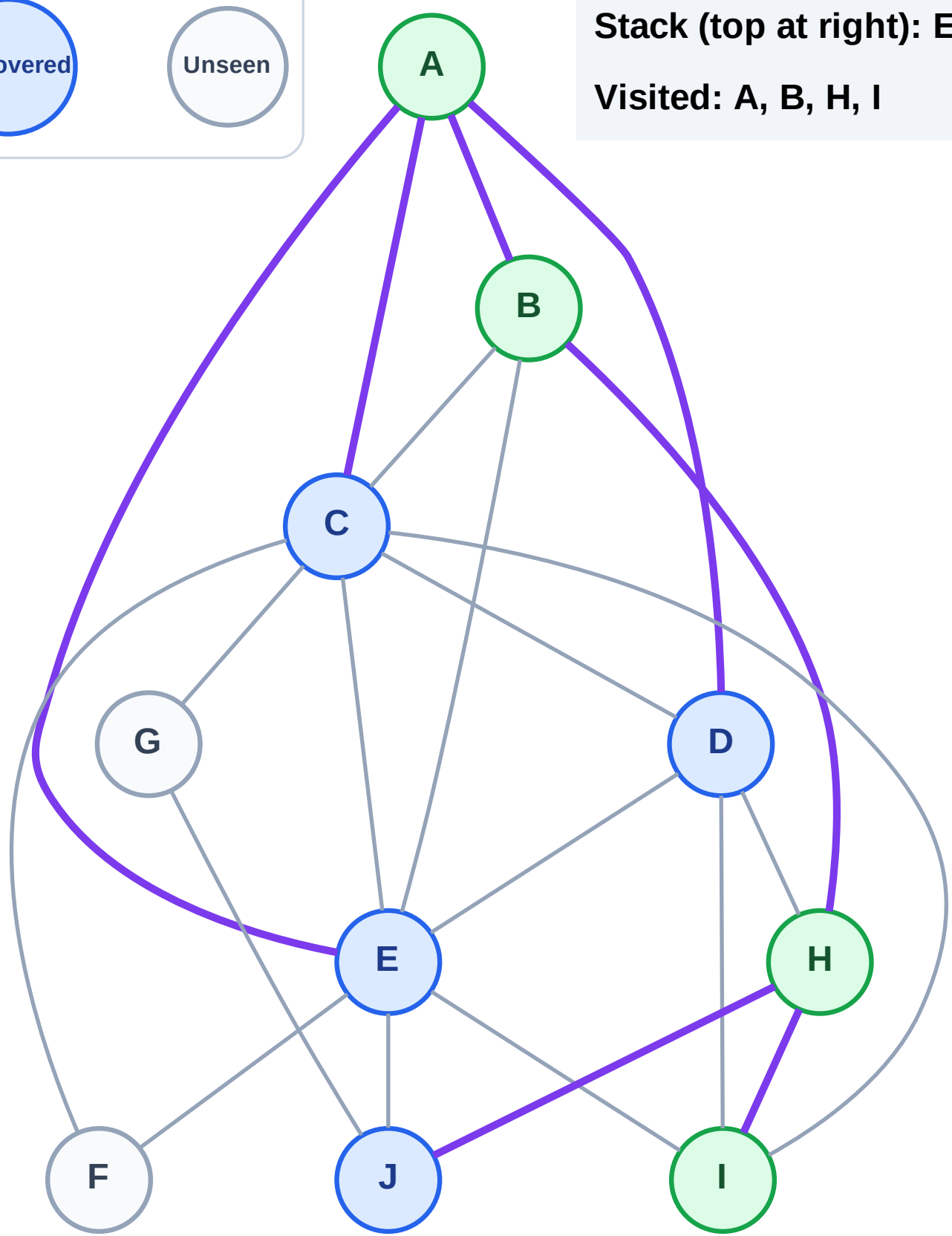
Step 9: No new neighbors from I

Legend



Stack (top at right): E | D | C | J

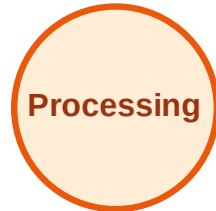
Visited: A, B, H, I



DFS Traversal

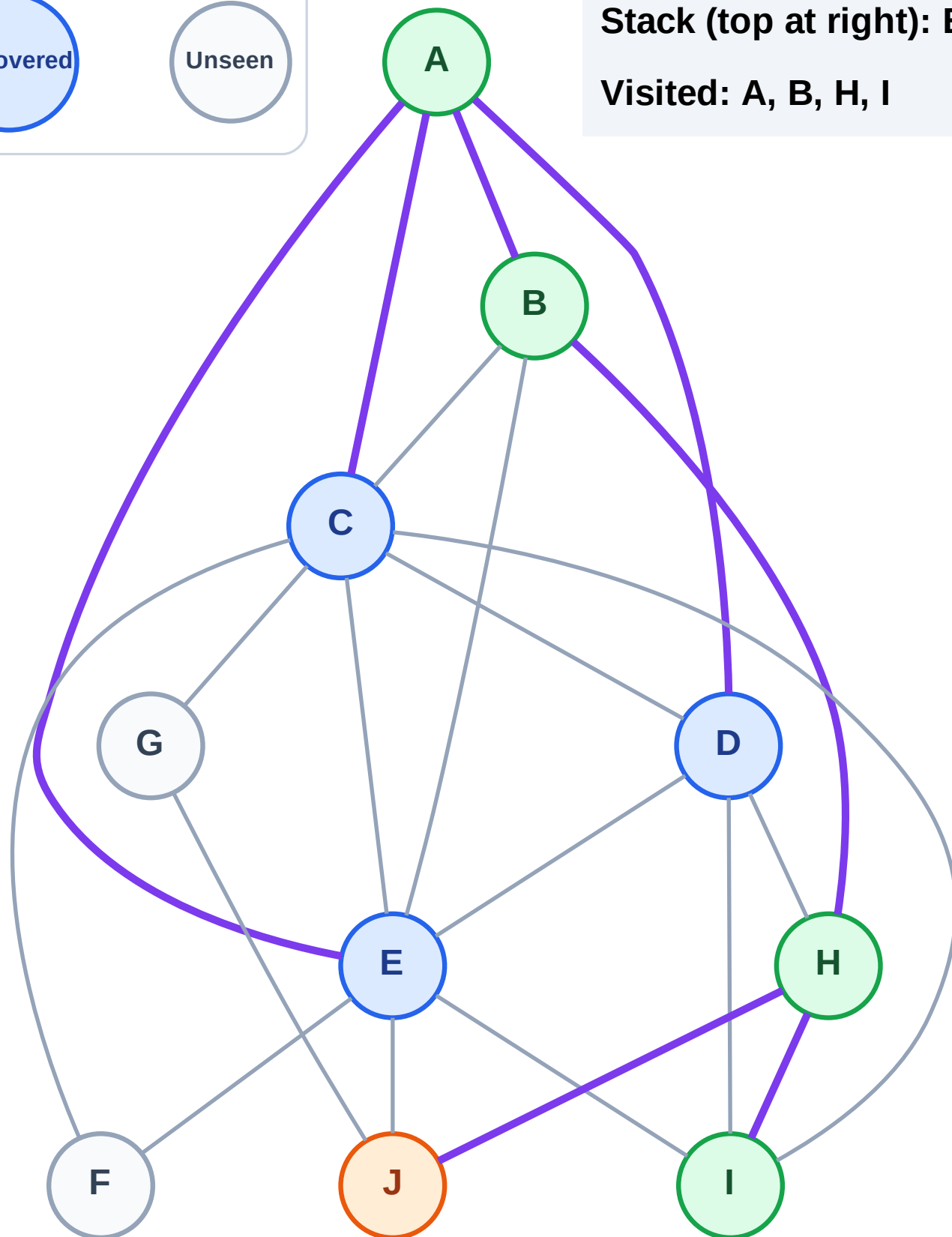
Step 10: Process node J

Legend



Stack (top at right): E | D | C

Visited: A, B, H, I



DFS Traversal

Step 11: Discover G from J

Legend

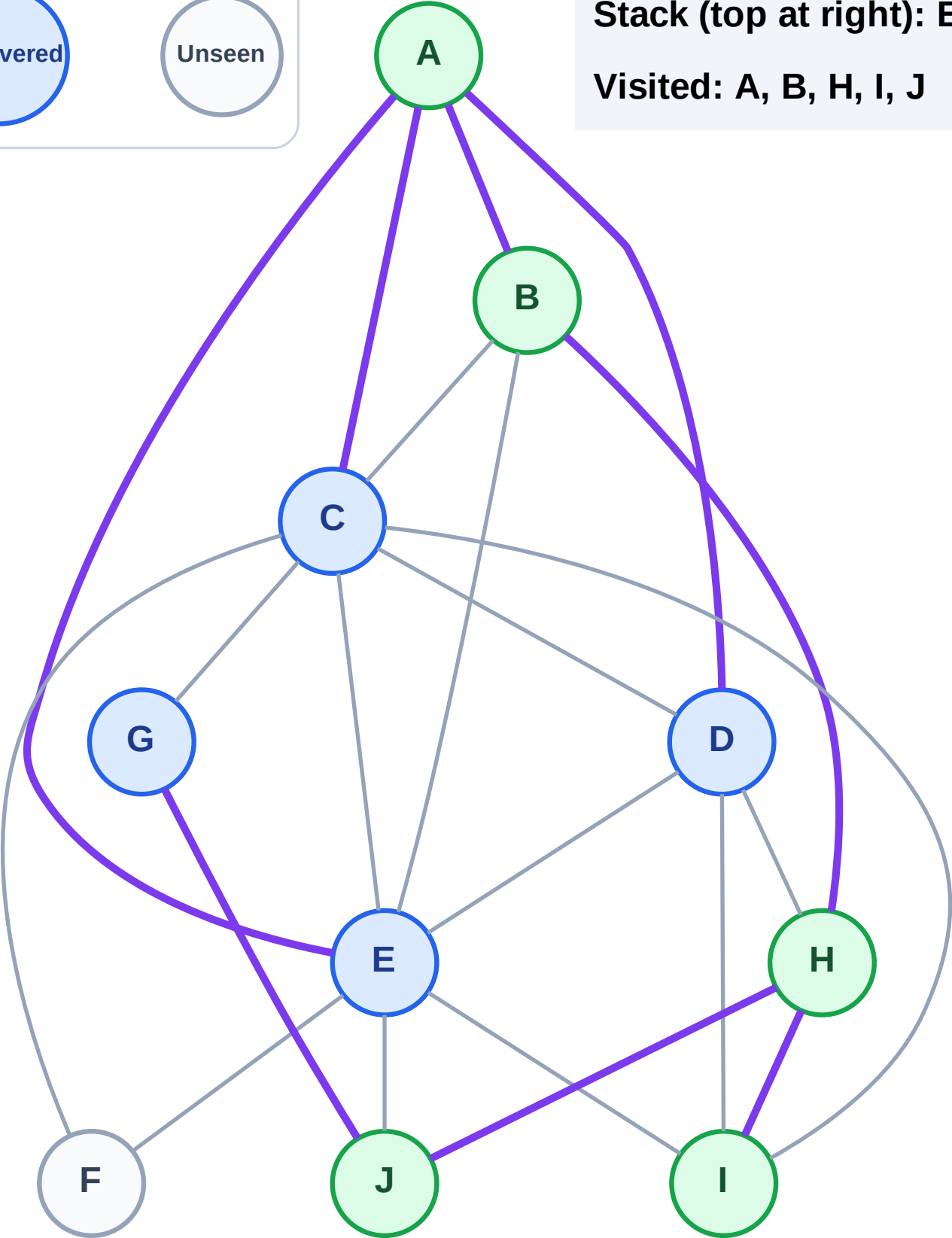
Complete

Processing

Discovered

Unseen

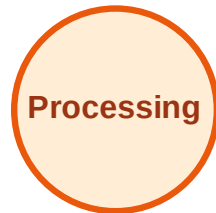
Stack (top at right): E | D | C | G
Visited: A, B, H, I, J



DFS Traversal

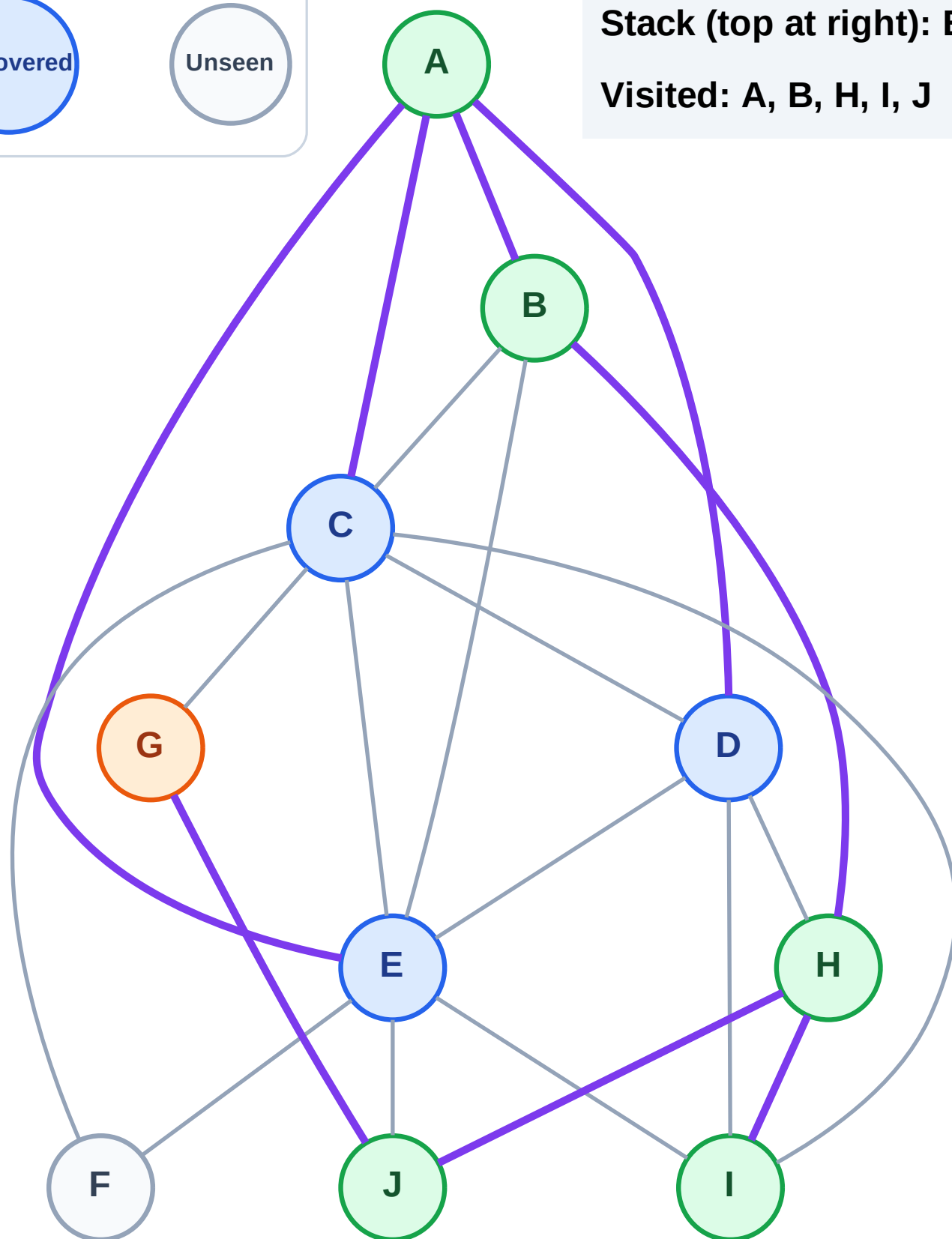
Step 12: Process node G

Legend



Stack (top at right): E | D | C

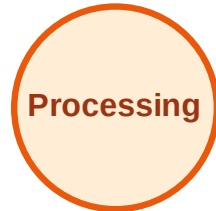
Visited: A, B, H, I, J



DFS Traversal

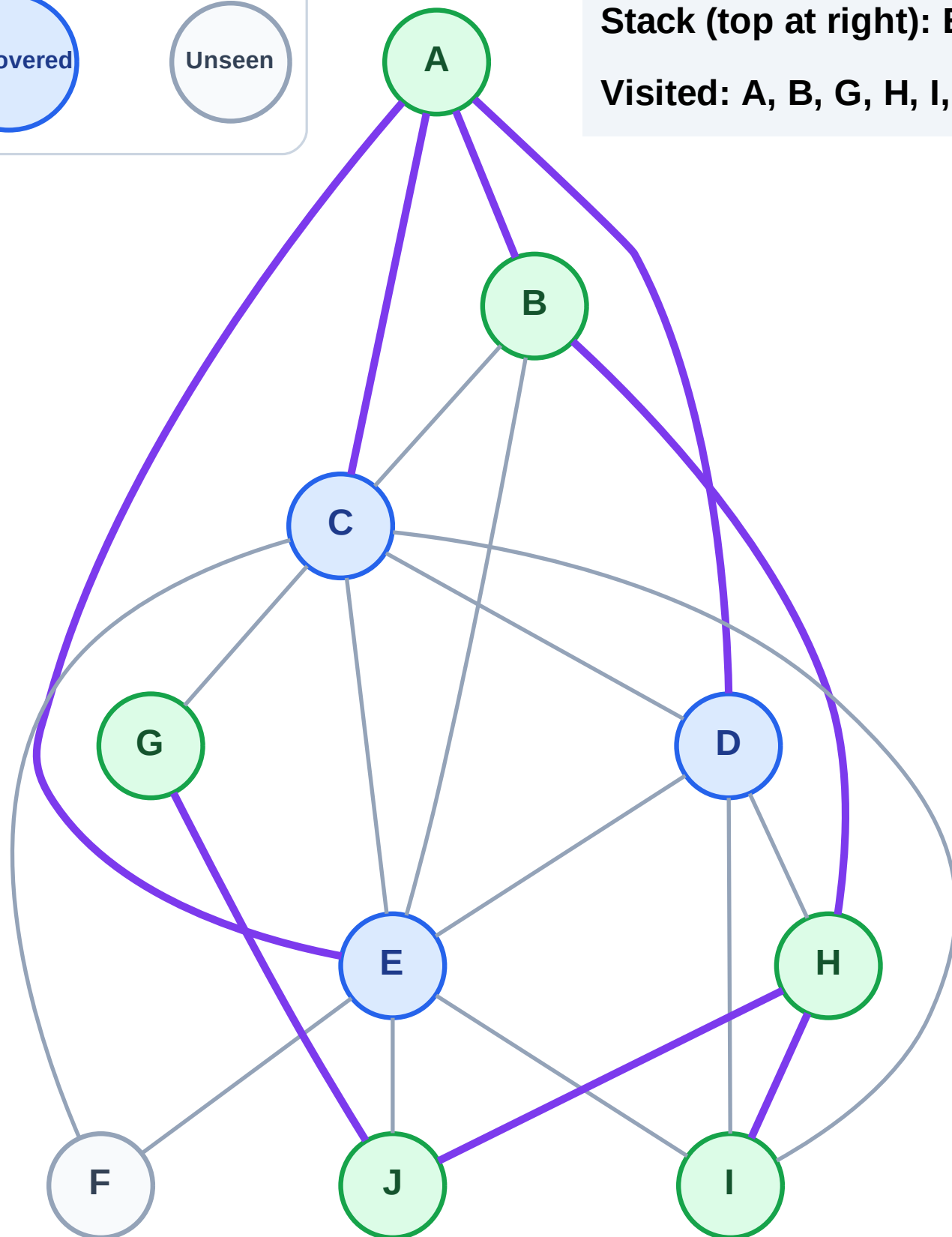
Step 13: No new neighbors from G

Legend



Stack (top at right): E | D | C

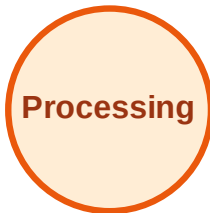
Visited: A, B, G, H, I, J



DFS Traversal

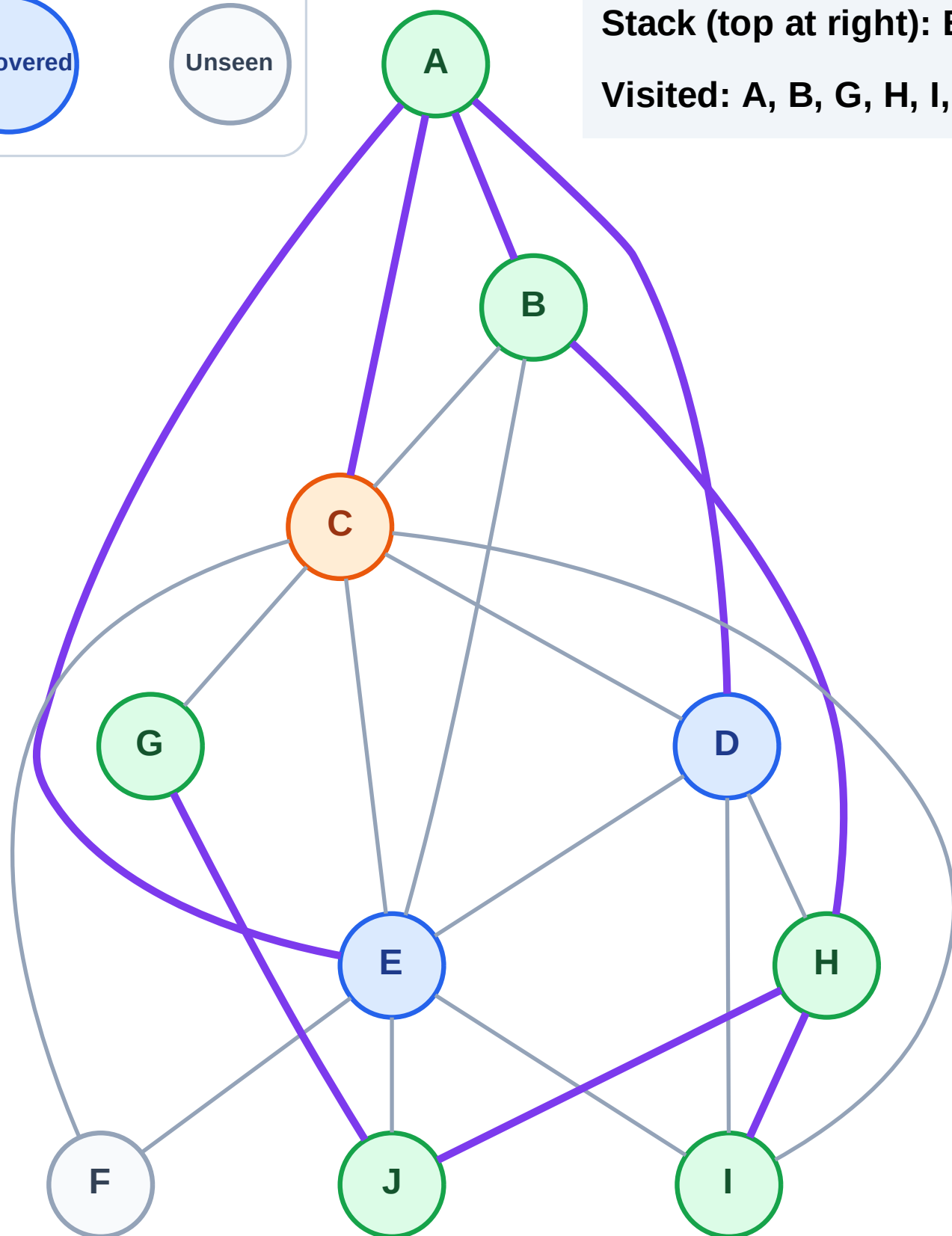
Step 14: Process node C

Legend



Stack (top at right): E | D

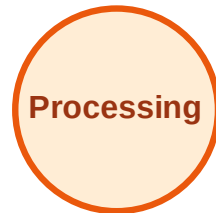
Visited: A, B, G, H, I, J



DFS Traversal

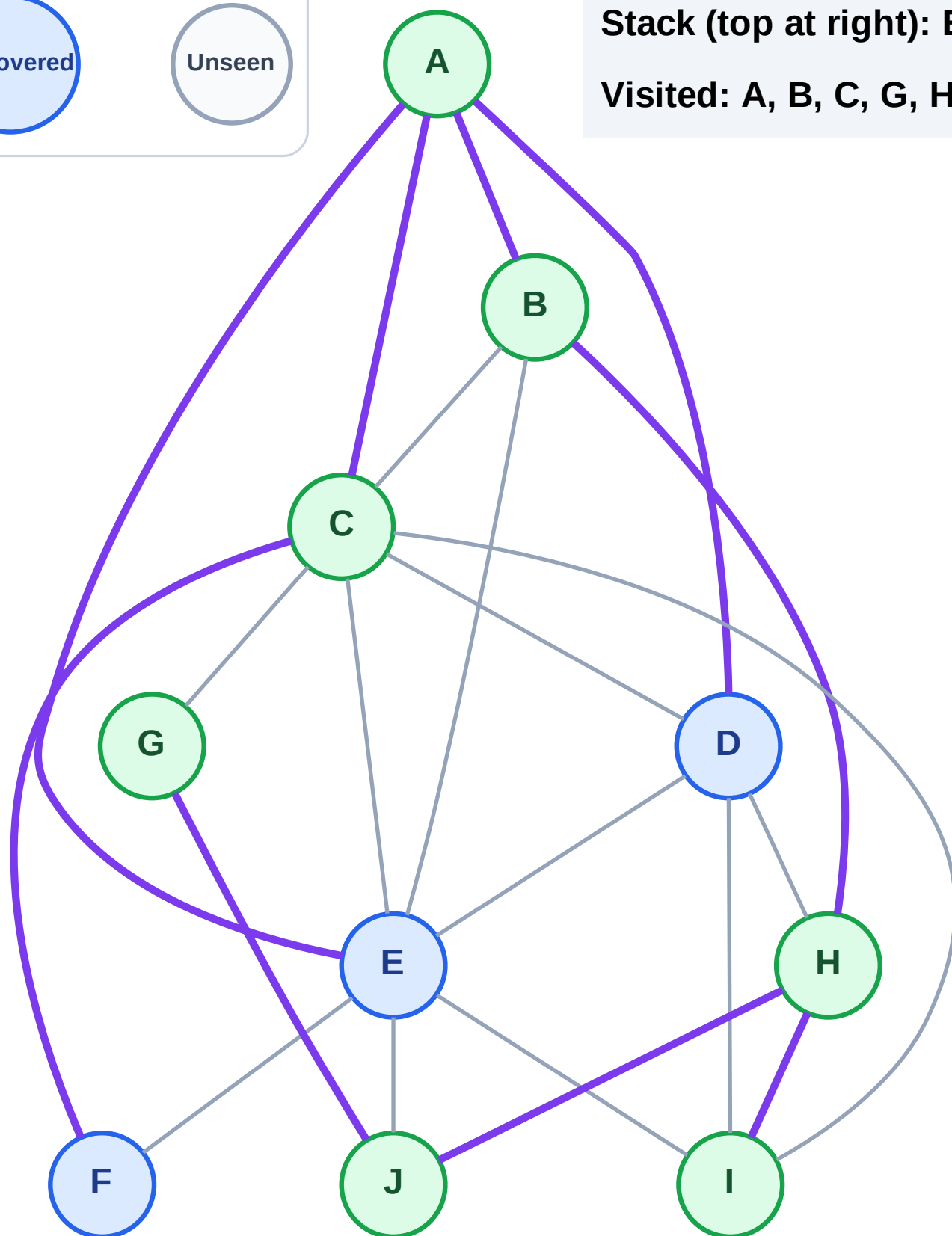
Step 15: Discover F from C

Legend



Stack (top at right): E | D | F

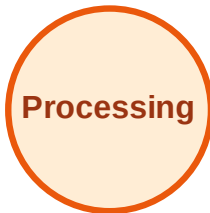
Visited: A, B, C, G, H, I, J



DFS Traversal

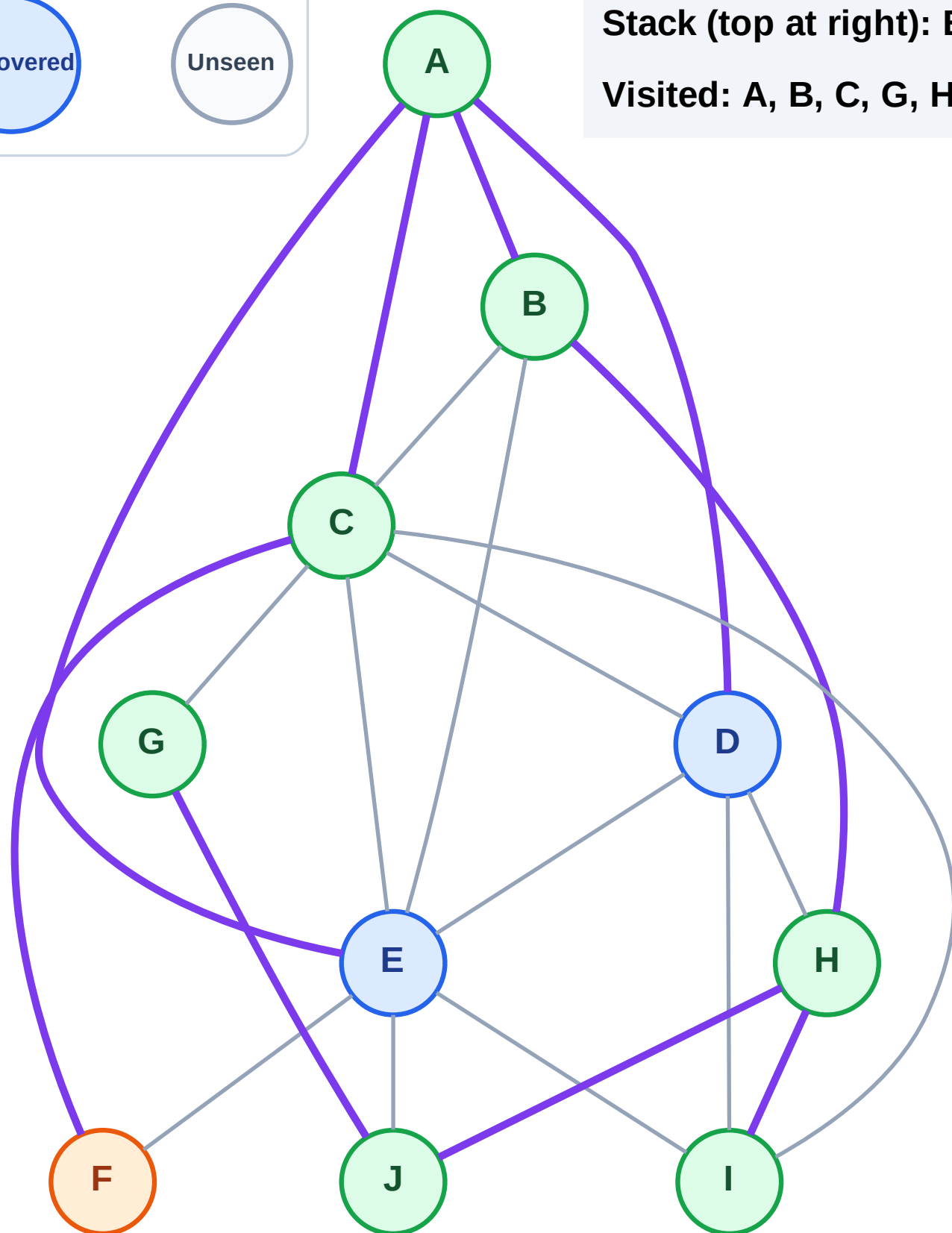
Step 16: Process node F

Legend



Stack (top at right): E | D

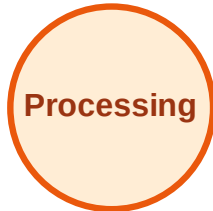
Visited: A, B, C, G, H, I, J



DFS Traversal

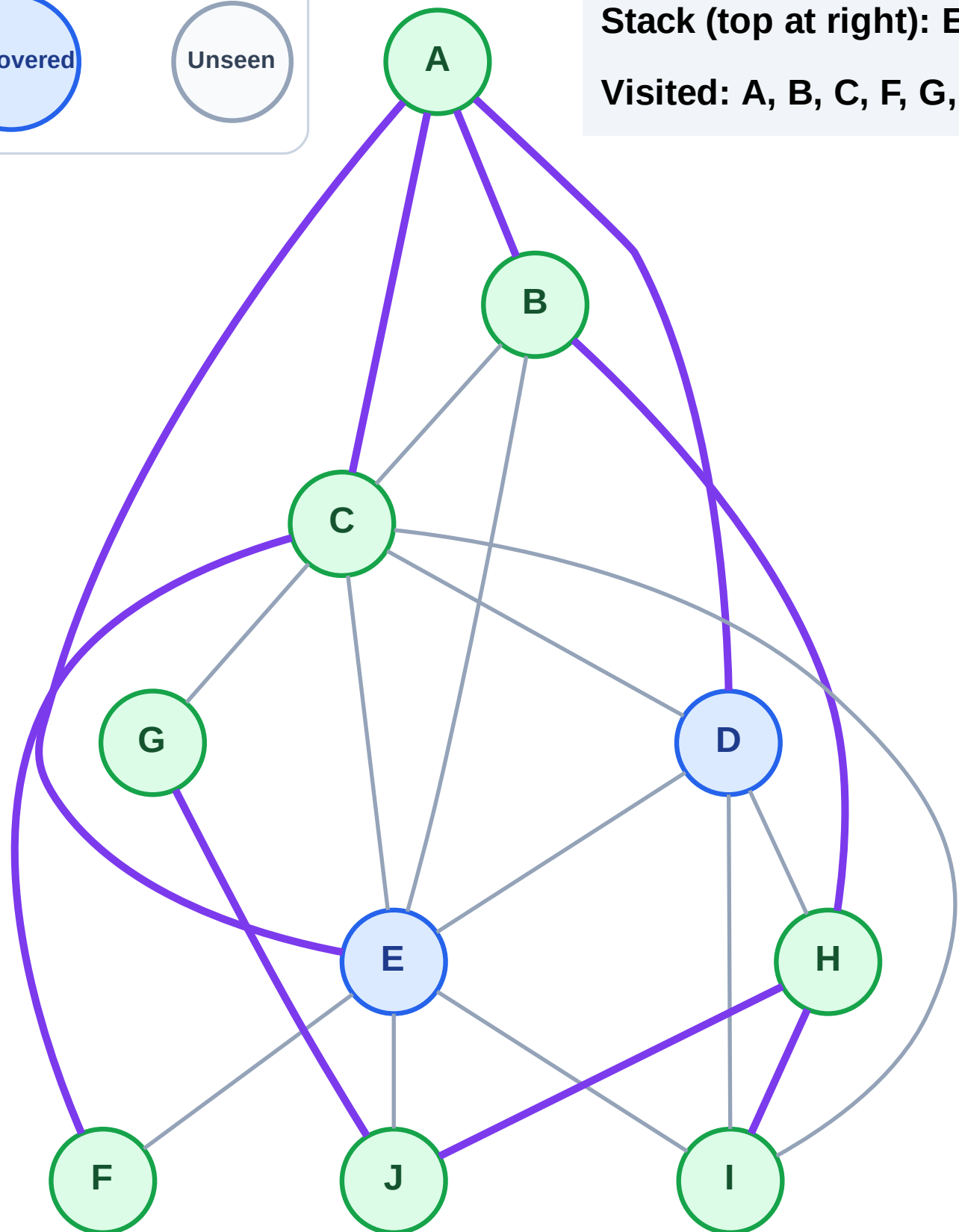
Step 17: No new neighbors from F

Legend



Stack (top at right): E | D

Visited: A, B, C, F, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

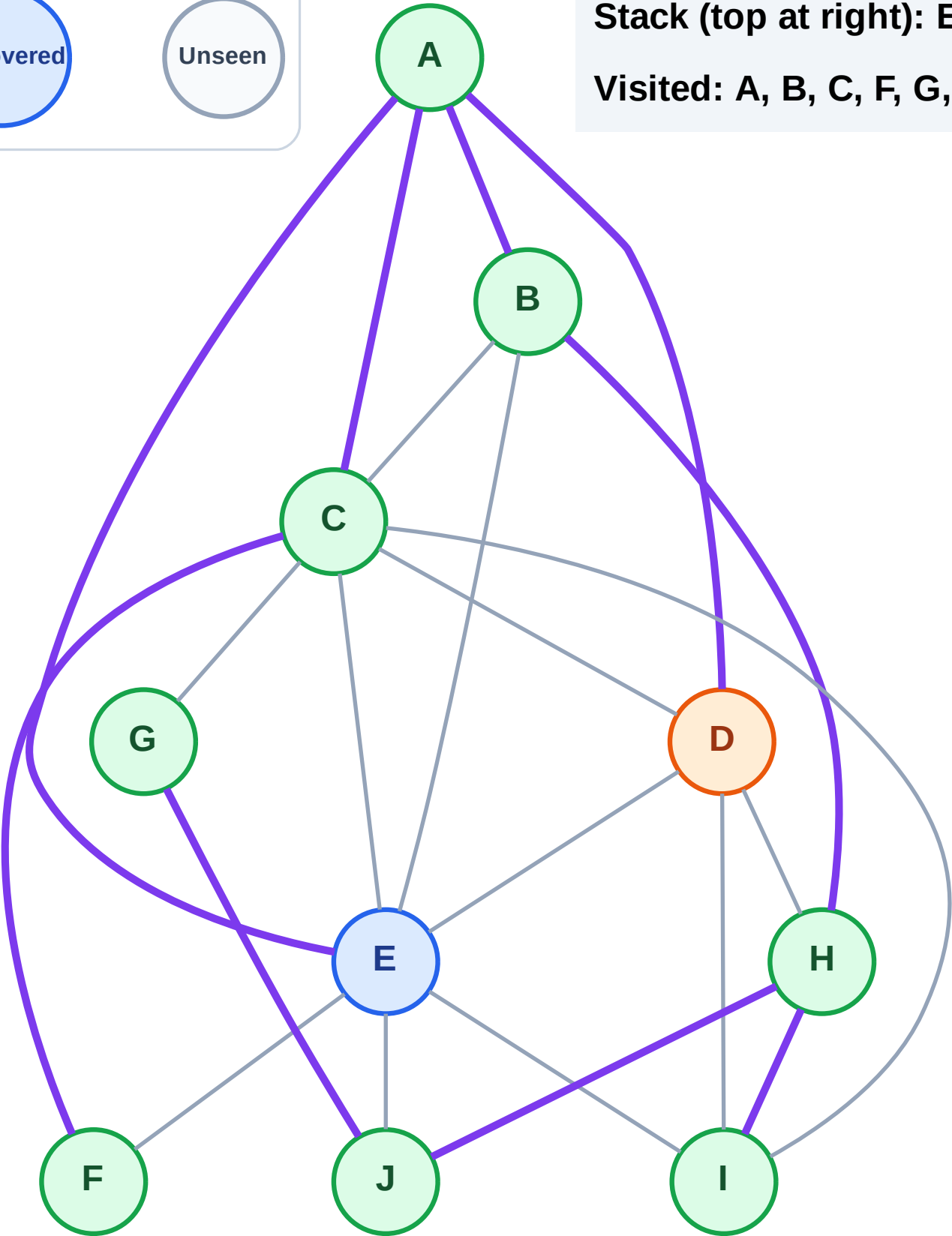
Complete

Processing

Discovered

Unseen

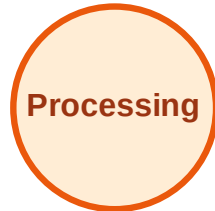
Stack (top at right): E
Visited: A, B, C, F, G, H, I, J



DFS Traversal

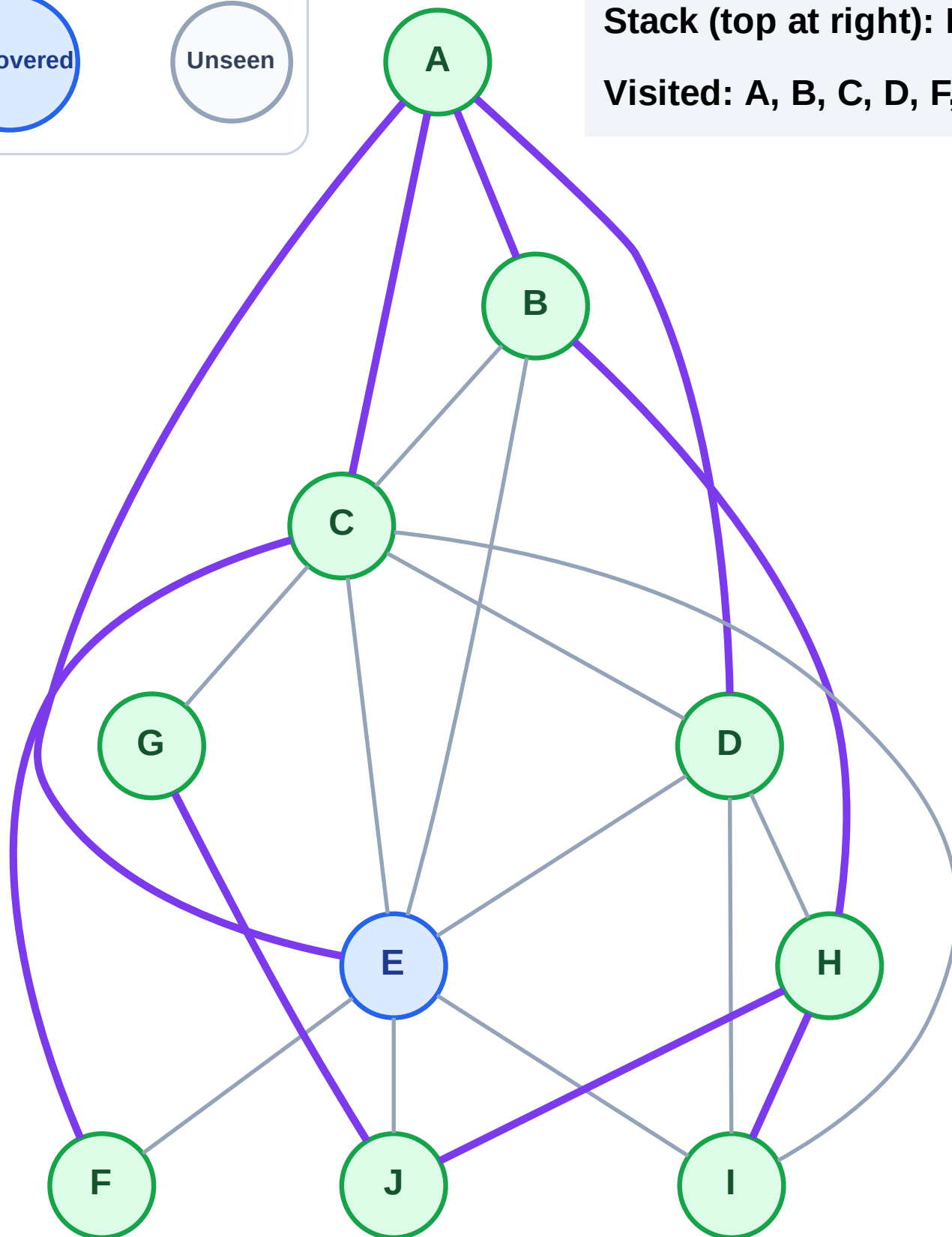
Step 19: No new neighbors from D

Legend



Stack (top at right): E

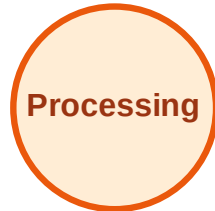
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

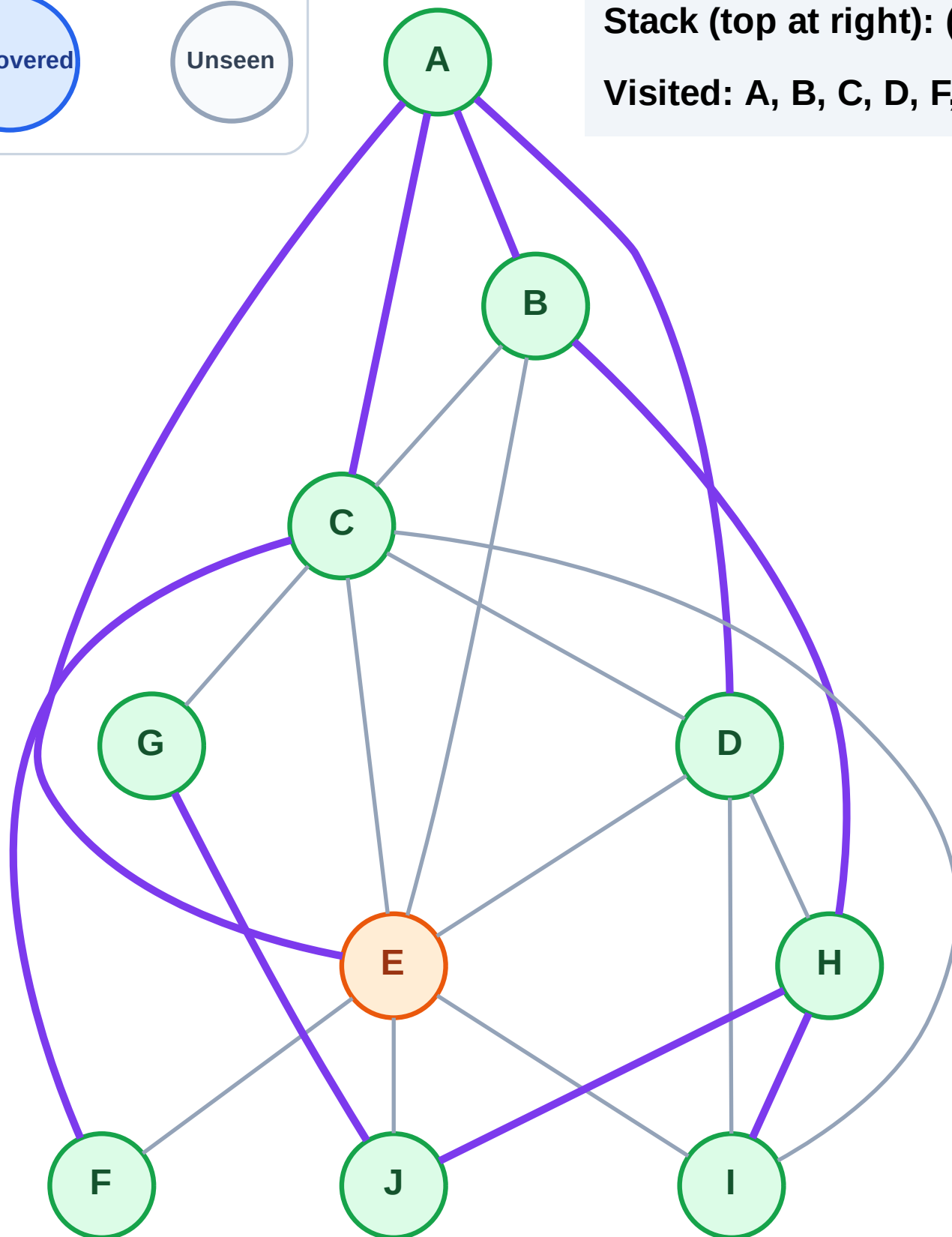
Step 20: Process node E

Legend



Stack (top at right): (empty)

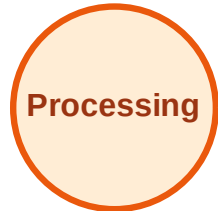
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

